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Study of Decoherence Effects on Polarization Entanglement Distribution through Optical Fibers

Tesi di Laurea Magistrale in
High Performance Computing Engineering

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Abstract

As quantum technologies progress toward distributed architectures, optical quantum links are becoming increasingly important for both interconnecting remote quantum processing nodes and enabling secure quantum communication. Polarization-encoded photonic qubits are particularly suited to fiber-based transmission, since polarization can be prepared, manipulated, and measured using well-established optical and quantum-optical components.

Among the applications considered in this thesis, particular attention is devoted to entanglement-based quantum key distribution, with the BBM92 protocol providing the main operational reference. In this context, the preservation of high-quality entanglement is directly related to the suitability of the distributed state for secure quantum communication.

This thesis develops a progressive and unified framework for studying the propagation of polarization-entangled photon pairs through optical fibers, connecting physical descriptions commonly treated separately at different levels of approximation. The main objective is to relate fiber and source parameters to the quality of the distributed entanglement and to the operational requirements of BBM92-based communication. To this purpose, the framework connects the physical parameters of the optical link with state-quality metrics and protocol-relevant quantities, particularly the quantum bit error rate (QBER). This provides a practical procedure for identifying operating regions of the physical parameter space in which the distributed state satisfies an adopted BBM92 QBER criterion, without assuming a universal concurrence threshold.

The modeling hierarchy starts from the residual polarization evolution after partial compensation. A fixed relative phase first distinguishes coherent changes in measured correlations from genuine entanglement degradation. The model is then extended to unresolved phase fluctuations through the coherence parameter $\mu = \langle e^{i\theta} \rangle$, with $C = |\mu|$, and to independent depolarizing effects in the two fiber arms through $\eta = (1 - p_A)(1 - p_B)$. Finally, these effective descriptions are connected to polarization-mode dispersion (PMD), where finite-bandwidth photons undergo frequency-dependent birefringent transformations and polarization mixedness emerges after tracing over unresolved spectral degrees of freedom.

The framework is implemented in a modular MATLAB simulator for quantum state preparation, fiber-channel composition, statistical averaging, spectral tracing, and polarization-correlation analysis. Analytical and numerical results are used to study how source properties, fiber parameters, PMD, and compensation conditions affect the surviving entanglement. The numerical framework is also structured to relate these physical parameters to QBER-based operational requirements and to identify the corresponding BBM92 operating regions.

The theoretical and numerical analysis is complemented by a preliminary laboratory study of a simplified BBM92-oriented polarization-entanglement platform, representing a first step toward evaluating practical conditions and requirements for entanglement-based quantum communication over optical fibers.

Keywords: Polarization Entanglement, Optical Fibers, BBM92, Quantum Channels, Decoherence, Depolarization.

Abstract in lingua italiana

DA INSERIRE QUANDO L'ABSTRACT È DEFINITIVO

Parole chiave: Entanglement di Polarizzazione, Fibre Ottiche, BBM92, Canali Quantistici, Decoerenza, Depolarizzazione.

Contents

Introduction

As quantum computing technologies continue to advance and quantum communication links become increasingly relevant for interconnecting remote quantum systems, the reliable transmission of quantum states between distant nodes is becoming an increasingly important technological requirement. Optical fibers represent a natural transmission medium for many such applications because of their compatibility with existing telecommunications infrastructure and their suitability for long-distance photonic links [1, 2].

Among the different photonic degrees of freedom, polarization provides a particularly convenient encoding for quantum information. Polarization-entangled photon pairs can be generated, manipulated, and measured using established optical components, making them attractive resources for entanglement distribution and entanglement-based quantum communication protocols [3]. Their transmission through real optical fibers, however, is affected by birefringence, polarization rotations, phase accumulation, temporal effects, and environmental fluctuations [4]. While deterministic local transformations can remain coherent and preserve entanglement, unresolved fluctuations or coupling to additional degrees of freedom can produce effective decoherence and degrade the polarization correlations accessible at the receiver [5, 6].

The preservation of polarization entanglement is particularly relevant to entanglement-based quantum key distribution. In the BBM92 protocol, an entangled photon pair is distributed between two distant parties, Alice and Bob, who independently choose between alternative polarization measurement bases to establish a secure key [3]. Fiber propagation and environmental perturbations can degrade the distributed correlations, increasing the quantum bit error rate (QBER) and ultimately limiting the suitability of the link for secure quantum communication.

BBM92 therefore provides the main operational reference for this thesis, whose objective is to investigate how source properties and fiber propagation conditions affect the quality of distributed polarization entanglement and its suitability for quantum communication. The reference configuration consists of an entangled-photon source connected to two spatially separated fiber arms, A and B, with one photon propagating through each arm and polarization used as the qubit degree of freedom. The work progressively connects the physical mechanisms affecting this configuration with their mathematical modeling, numerical simulation, and operational consequences.

Chapter 1 establishes the physical and mathematical framework required for this analysis. It introduces the description of polarization qubits and bipartite states, the density-operator formalism, reduced states, local measurements, and polarization correlations. The chapter also presents the quantities used throughout the thesis to characterize state quality, coherence, entanglement, and nonlocality. Particular attention is devoted to concurrence as the principal entanglement measure adopted in the numerical analysis. A brief description of the BBM92 protocol and of its main quantities of interest is also provided, establishing the connection between the quantum-state description and the operational perspective used in the subsequent chapters.

Chapter 2 describes how optical fiber propagation modifies the polarization state through a progressive hierarchy of models with increasing physical detail. A central distinction is made between coherent polarization evolution, which can modify measured correlations without necessarily reducing purity or entanglement, and effective decoherence arising from unresolved fluctuations or coupling to unobserved degrees of freedom. Reduced models are first used to isolate coherent phase evolution, statistical dephasing, and local depolarizing effects in the two propagation arms, providing analytically interpretable descriptions of the different degradation mechanisms. The analysis is then extended to a frequency-dependent treatment of polarization-mode dispersion (PMD) [7–9], connecting fiber birefringence and the spectral properties of the photon pair to the effective degradation of polarization entanglement. The resulting hierarchy therefore provides a continuous path from simplified effective channels to a physical description of frequency-dependent fiber propagation.

Chapter 3 implements the developed models in a modular MATLAB simulation framework and investigates through numerical experiments how source properties, fiber parameters, PMD, and compensation conditions affect the distributed entanglement. The analysis aims to identify regions of parameter space in which a prescribed entanglement quality is maintained, assess the validity of reduced models against the physical frequency-dependent description, and relate the resulting entanglement degradation to BBM92-oriented operational requirements.

Chapter **??** complements the theoretical and numerical work with a preliminary laboratory study of polarization-entanglement distribution. A simplified two-arm platform based on entangled-photon propagation, polarization control, single-photon detection, and coincidence measurements is considered. The objective of this activity is to provide a first laboratory approach to the measurement conditions and practical requirements relevant to BBM92-oriented quantum communication over optical fibers. At the present stage, the experimental activity is therefore intended as an initial exploration of the physical platform rather than as a complete BBM92 implementation.

Finally, Chapter **??** summarizes the main theoretical, numerical, and experimental results and discusses the possible extensions of the developed framework. Particular attention is given to the connection between physical fiber-channel modeling, the numerical identification of entanglement-quality regions, and their operational interpretation in the context of entanglement-based quantum communication.

1 | Mathematical and Physical Framework

This chapter introduces the physical representation and mathematical tools used throughout the thesis to describe polarization-entangled photon pairs and their evolution through optical-fiber channels. The treatment is formulated within the density-operator framework of quantum information theory, which provides a common description for pure states, statistical mixtures, local measurements, and noisy quantum evolution [10].

The chapter first defines the bipartite polarization system and the reference Bell state. The density-operator and reduced-state formalisms are then introduced, followed by the description of two-photon polarization measurements and correlations. The main quantities used to characterize the propagated quantum state are subsequently defined, with particular attention to purity, fidelity, concurrence, tangle, and entanglement of formation.

Finally, the chapter introduces the BBM92 entanglement-based quantum key distribution protocol as the main operational reference of this work. Its measurement scheme and principal quantities of interest, including the quantum bit error rate and secret-key conditions, are presented in order to establish the connection between the quality of the distributed entangled state and its suitability for secure quantum communication. This provides the operational basis for interpreting the fiber-propagation models and numerical results developed in the following chapters.

1.1. Bipartite Polarization System

The physical system considered in this work consists of a pair of polarization-entangled photons propagating along two spatially separated optical paths, denoted by A and B. In the experimental configuration considered later in this thesis, the photon pairs are produced through high-fidelity spontaneous parametric down-conversion (SPDC).

The quantum information carried by each photon is encoded in its polarization degree of freedom. Horizontal and vertical polarization are identified with the computational basis

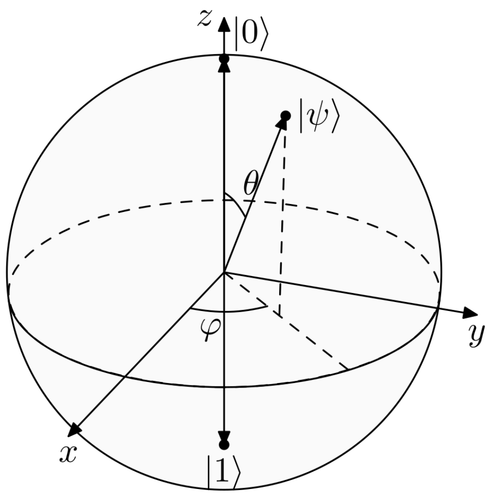


Figure 1.1: Bloch-sphere representation of a single polarization qubit. The horizontal and vertical polarization states correspond to $|H\rangle \equiv |0\rangle$ and $|V\rangle \equiv |1\rangle$.

according to

$$|H\rangle \equiv |0\rangle, \qquad |V\rangle \equiv |1\rangle. \tag{1.1}$$

Each photon therefore defines a two-dimensional Hilbert space,

$$\mathcal{H}_A \simeq \mathbb{C}^2, \qquad \mathcal{H}_B \simeq \mathbb{C}^2. \tag{1.2}$$

The Bloch sphere provides a geometric representation of the state of a single qubit, as illustrated in Fig. 1.1. Pure states are represented by points on the surface of the sphere, while the polar angle and azimuthal angle determine the relative amplitudes and phase of the two basis-state components. Coherent superpositions occupy the remaining points on the sphere, with equal-weight superpositions lying on the equator.

The joint system is described in the tensor-product Hilbert space

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B \simeq \mathbb{C}^2 \otimes \mathbb{C}^2, \tag{1.3}$$

which has dimension four.

The ordered computational basis adopted throughout the thesis is

$$\mathcal{B}_{AB}^{01} = \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}, \tag{1.4}$$

where the first entry refers to subsystem A and the second to subsystem B. In polarization notation, the same basis is

$$\mathcal{B}_{AB}^{HV} = \{|HH\rangle, |HV\rangle, |VH\rangle, |VV\rangle\}. \tag{1.5}$$

Consequently, pure states of the complete system are represented by four-component complex vectors and operators acting on both photons by 4×4 matrices. This basis convention is used consistently in the analytical derivations and in the MATLAB implementation.

1.2. Reference Entangled State

The principal input state considered in this work is the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right), \tag{1.6}$$

or equivalently,

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|HV\rangle + |VH\rangle\right). \tag{1.7}$$

In the ordered basis of Eq. (1.4), its vector representation is

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}. \tag{1.8}$$

For completeness, the four Bell states are

$$|\Psi^\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}, \qquad |\Phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}. \tag{1.9}$$

The Bell states are examples of bipartite entangled states. A bipartite pure state is separable if it can be written as a product of independent subsystem states,

$$|\psi_{AB}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle. \tag{1.10}$$

If such a factorization is not possible, the state is entangled [10]. Entanglement therefore describes correlations belonging to the joint quantum state that cannot be reduced to independent descriptions of the two subsystems.

The state $|\Psi^+\rangle$ is maximally entangled and therefore provides both a physically relevant resource and a convenient analytical benchmark for studying fiber-induced transformations. In this case, the relevant information is not contained in the state of either photon individually, but in the correlations shared between the two spatially separated subsystems.

This makes $|\Psi^+\rangle$ particularly suitable for investigating fiber propagation: coherent local transformations may modify the observed correlation pattern without changing the amount of entanglement, whereas decoherence mechanisms can progressively reduce the shared correlations and eventually degrade the distributed entanglement. Monitoring the evolution of $|\Psi^+\rangle$ therefore provides a direct way to distinguish reversible polarization transformations from genuine loss of entanglement quality in the fiber link.

Although the theoretical and numerical frameworks apply to arbitrary two-qubit states, $|\Psi^+\rangle$ is used as the reference state throughout most of the simulations and laboratory analysis.

1.3. Density-Operator Description

A pure quantum state can be represented either by its state vector $|\psi\rangle$ or by the corresponding density operator

$$\rho = |\psi\rangle\langle\psi|. \tag{1.11}$$

For a bipartite system, the composite state is described by the density operator ρ_{AB}. If the global bipartite state is pure, then

$$\rho_{AB} = |\psi_{AB}\rangle\langle\psi_{AB}|. \tag{1.12}$$

The density-operator formulation is equivalent to the state-vector description for pure states, but it extends naturally to situations in which the system is mixed.

A general mixed state may be expressed as

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|, \qquad p_k \geq 0, \qquad \sum_k p_k = 1. \tag{1.13}$$

Such states arise naturally when different physical realizations are averaged statistically or when degrees of freedom correlated with the system are not resolved experimentally. Importantly, a density operator does not in general possess a unique ensemble decomposition; different statistical ensembles may represent the same physical state [10].

The density-operator formalism is adopted as the common representation throughout the work and the simulator.

1.3.1. Expectation Values

For a pure state, the expectation value of an observable O is

$$\langle O\rangle = \langle\psi|O|\psi\rangle. \tag{1.14}$$

In density-operator notation, the corresponding general expression is

$$\langle O\rangle = \mathrm{Tr}(\rho O). \tag{1.15}$$

For $\rho = |\psi\rangle\langle\psi|$, Eq. (1.15) reduces directly to the pure-state expression. The same equation remains valid for mixed states and therefore provides the basis for the evaluation of all observables used in this work.

1.3.2. Reduced Density Operators

For a bipartite state ρ_{AB}, the state accessible locally to either subsystem is obtained through the partial trace,

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}), \qquad \rho_B = \mathrm{Tr}_A(\rho_{AB}). \tag{1.16}$$

Mathematically, tracing out subsystem B means summing over a complete basis of $\mathcal{H}_B$, thereby eliminating the degrees of freedom associated with B and producing the unique operator on $\mathcal{H}_A$ that reproduces all local measurement statistics on A.

For example, tracing over subsystem B gives

$$\rho_A = \sum_{j=0}^{1} (I_A \otimes \langle j|)\, \rho_{AB}\, (I_A \otimes |j\rangle)\,. \tag{1.17}$$

For an observable O_A, characterizing subsystem A

$$\mathrm{Tr}\left[\rho_{AB}\left(O_A \otimes I_B\right)\right] = \mathrm{Tr}(\rho_A O_A). \tag{1.18}$$

For the Bell state $|\Psi+\rangle$ of Eq. (1.6), the two reduced density operators are

$$\rho_A = \rho_B = \frac{I_2}{2}. \tag{1.19}$$

Thus, each photon is locally maximally mixed even though the complete two-photon state is pure and maximally entangled. The information distinguishing the Bell state from an uncorrelated local mixture is therefore contained in the joint correlations rather than in the individual reduced states.

1.3.3. Bloch Representation

Any single-qubit density operator can be written as

$$\rho = \frac{1}{2}\left(I + \mathbf{r}\cdot\boldsymbol{\sigma}\right), \tag{1.20}$$

where

$$\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$$

is the vector of Pauli matrices and $\mathbf{r} \in \mathbb{R}^3$ is the Bloch vector.

Physical states satisfy

$$\|\mathbf{r}\| \leq 1. \tag{1.21}$$

Pure states satisfy $\|\mathbf{r}\| = 1$ and lie on the surface of the Bloch sphere, whereas mixed states satisfy $\|\mathbf{r}\| < 1$. The maximally mixed state corresponds to

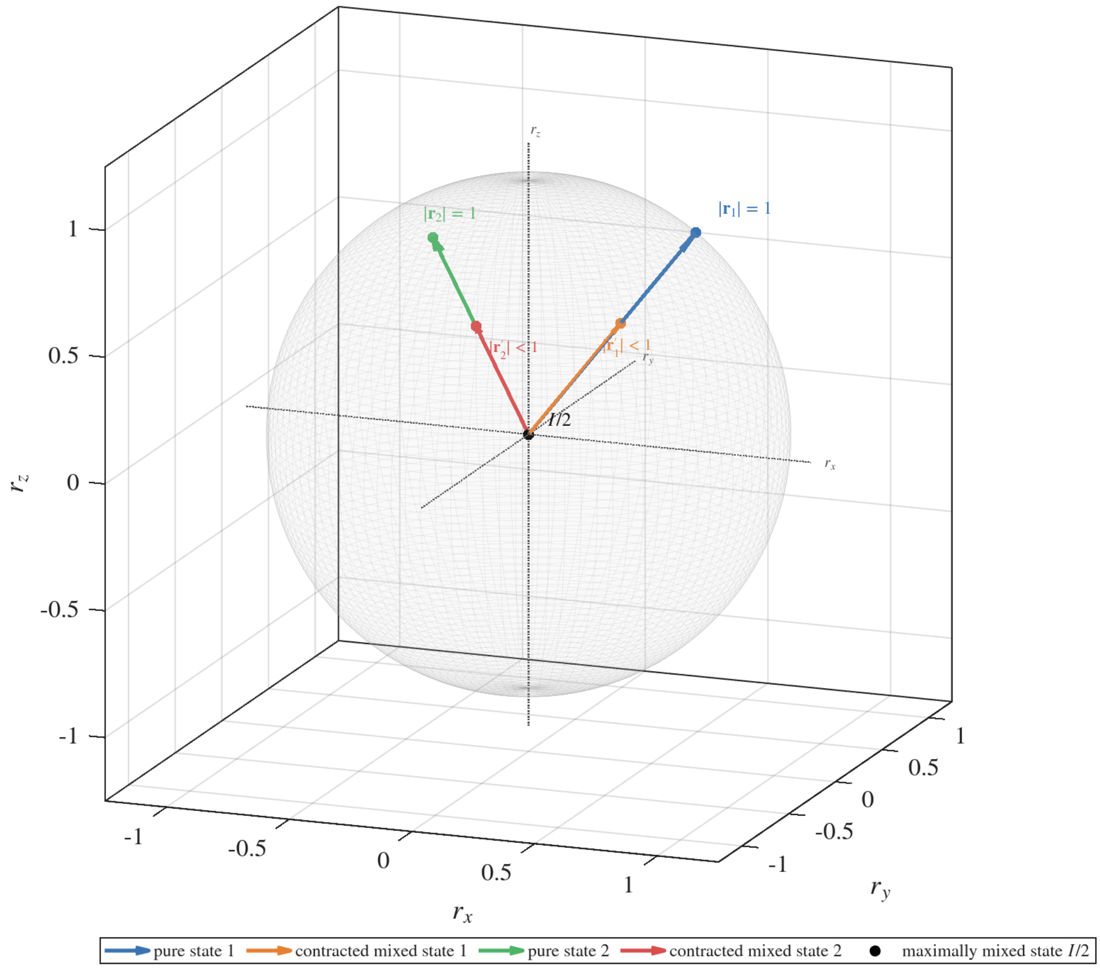


Figure 1.2: Bloch-sphere representation of pure and mixed single-qubit states. Pure states satisfy $\|\mathbf{r}\| = 1$, whereas mixed states are represented by vectors inside the Bloch sphere. The maximally mixed state corresponds to the origin.

$$\mathbf{r} = 0, \qquad \rho = \frac{I_2}{2}.$$

Unitary transformations rotate the Bloch vector while preserving its norm. Effective decohering processes may instead reduce its magnitude, corresponding to motion toward the interior of the Bloch ball.

1.4. Quantum Evolution and Quantum Channels

The evolution of the two polarization qubits can occur either coherently or through an effective noisy process. These two regimes are naturally distinguished within the density-operator formalism.

1.4.1. Local Unitary Evolution

When each photon undergoes a coherent polarization transformation, its evolution is represented by a unitary operator acting on the corresponding subsystem.

If U_A and U_B denote the transformations acting on arms A and B, respectively, the pure state evolves according to

$$|\psi_{\text{out}}\rangle = (U_A \otimes U_B)|\psi_{\text{in}}\rangle. \tag{1.22}$$

At the density-operator level,

$$\rho_{\text{out}} = (U_A \otimes U_B)\rho_{\text{in}}(U_A^\dagger \otimes U_B^\dagger). \tag{1.23}$$

Local unitary evolution preserves global purity and bipartite entanglement. It may nevertheless modify the correlations observed with respect to fixed laboratory measurement axes. This distinction between a coherent transformation and genuine entanglement degradation is central to the fiber models developed in Chapter 2.

1.4.2. Quantum Channels

A more general transformation of a quantum state is represented by a quantum channel,

$$\mathcal{E} : \rho \mapsto \mathcal{E}(\rho), \tag{1.24}$$

which is a completely positive and trace-preserving linear map.

A general channel may be written in operator-sum form as [10, 11]

$$\mathcal{E}(\rho) = \sum_\mu K_\mu \rho K_\mu^\dagger, \qquad \sum_\mu K_\mu^\dagger K_\mu = I, \tag{1.25}$$

where the operators K_μ are the Kraus operators associated with the map.

Unitary evolution is recovered as the special case containing a single Kraus operator $K = U$. More general channels allow the reduced state to become mixed and are therefore required to describe decoherence, depolarization, and other effective noise mechanisms.

The specific channel models used to represent optical-fiber propagation are introduced in

Chapter 2.

1.5. Polarization Correlations and Measurements

For a bipartite polarization state, the physically relevant information is not contained only in the statistics observed separately on arms A and B, but also in how the outcomes of measurements performed on the two photons are related.

A correlation quantifies this statistical relation. If dichotomic polarization measurements are performed on the two arms, with outcomes $a, b \in \{+1, -1\}$, their correlation is defined as the expectation value of the product of the two outcomes. A positive correlation indicates a tendency to obtain equal outcomes, whereas a negative correlation indicates a tendency to obtain opposite outcomes. A vanishing correlation means that the positive and negative contributions balance for the chosen pair of measurement axes.

For measurements described by Pauli operators, these joint statistics are conveniently represented through tensor products of local observables.

1.5.1. Correlation Tensor

For a two-qubit state ρ_{AB}, the correlation tensor is defined by

$$T_{ij} = \mathrm{Tr}\left[\rho_{AB}\left(\sigma_i \otimes \sigma_j\right)\right], \qquad i, j \in \{x, y, z\}. \tag{1.26}$$

Each component T_{ij} is therefore the expectation value of the product of the outcomes obtained when measuring polarization along axis i on photon A and along axis j on photon B. Since the corresponding measurement outcomes are assigned the values ± 1, one has

$$-1 \leq T_{ij} \leq 1. \tag{1.27}$$

In particular, $T_{ij} = +1$ corresponds to perfect correlation, $T_{ij} = -1$ to perfect anticorrelation, while intermediate values quantify partial correlation for the selected pair of measurement settings.

Equivalently, denoting by $P_{ab}^{(i,j)}$ the joint probability of obtaining outcomes $a, b \in \{+1, -1\}$ when measuring along axes i and j,

$$T_{ij} = P_{++}^{(i,j)} + P_{--}^{(i,j)} - P_{+-}^{(i,j)} - P_{-+}^{(i,j)}. \tag{1.28}$$

This expression makes explicit that the correlation is determined by joint measurement statistics rather than by the local probabilities associated with either photon separately.

The tensor $T \in \mathbb{R}^{3\times 3}$ therefore collects the nine bipartite Pauli correlations. A single component $T_{ij} = 0$ does not in general imply that the two subsystems are statistically independent, since correlations may still be present along other pairs of measurement directions.

Introducing $\sigma_0 = I$, a general two-qubit state can be expanded as

$$\rho_{AB} = \frac{1}{4} \sum_{\mu,\nu \in \{0,x,y,z\}} S_{\mu\nu} \sigma_\mu \otimes \sigma_\nu, \tag{1.29}$$

where

$$S_{\mu\nu} = \mathrm{Tr}\left[\rho_{AB}\left(\sigma_\mu \otimes \sigma_\nu\right)\right]. \tag{1.30}$$

The quantities $S_{\mu\nu}$ correspond to generalized Stokes parameters. The terms S_{i0} and S_{0j} describe the local Bloch vectors, whereas the 3×3 block S_{ij} coincides with the correlation tensor T.

This representation is particularly useful in polarization experiments because the density operator can be reconstructed from expectation values of local Pauli measurements, as in standard two-qubit quantum-state tomography [12].

The reconstruction follows from the orthogonality relation

$$\mathrm{Tr}\left[(\sigma_i \otimes \sigma_j)(\sigma_k \otimes \sigma_l)\right] = 4\,\delta_{ik}\delta_{jl}, \qquad i,j,k,l \in \{0,x,y,z\}. \tag{1.31}$$

Experimental interpretation. In polarization-entangled photon experiments, the joint probabilities entering Eq. (1.28) are estimated from coincidence counts between detectors associated with the corresponding outcomes on arms A and B. A coincidence identifies two detections occurring within a prescribed temporal window and is interpreted as originating from a single emission of photon pair. After normalization, the measured coincidence counts provide experimental estimates of the joint probabilities required to evaluate the polarization correlations. The practical implementation of this procedure is discussed in chapter **??** about experimental activity.

1.5.2. Arbitrary Local Measurement Axes

A projective qubit measurement along a unit vector $\mathbf{n} \in \mathbb{R}^3$ is represented by the observable

$$\sigma(\mathbf{n}) = \mathbf{n} \cdot \boldsymbol{\sigma} = n_x \sigma_x + n_y \sigma_y + n_z \sigma_z, \qquad \|\mathbf{n}\| = 1. \tag{1.32}$$

The normalization condition ensures that the observable has eigenvalues ± 1.

For measurement directions $\mathbf{a}$ and $\mathbf{b}$ on subsystems A and B, respectively, the two-photon correlation is

$$E(\mathbf{a}, \mathbf{b}) = \mathrm{Tr}\left[\rho_{AB} \left(\sigma(\mathbf{a}) \otimes \sigma(\mathbf{b})\right)\right]. \tag{1.33}$$

Using Eq. (1.26), this becomes

$$E(\mathbf{a}, \mathbf{b}) = \mathbf{a}^{\mathsf{T}} T \mathbf{b}. \tag{1.34}$$

The correlation tensor therefore determines the expectation value associated with any pair of local Pauli measurement directions.

Experimentally, measurements along arbitrary axes can be realized by applying appropriate local polarization transformations before a fixed polarization analysis stage. This provides the connection between the abstract operator description and the wave-plate and polarization-controller settings used in the laboratory.

1.6. Quantitative Characterization of the State

No single scalar quantity completely characterizes a bipartite quantum state. Different measures are therefore used in this work to distinguish mixedness, proximity to a target state and entanglement.

1.6.1. Purity

The purity of a density operator is defined as

$$\gamma(\rho) = \mathrm{Tr}(\rho^2). \tag{1.35}$$

If ρ acts on a d-dimensional Hilbert space,

$$\frac{1}{d} \leq \gamma(\rho) \leq 1. \tag{1.36}$$

The upper bound corresponds to a pure state, whereas the lower bound is attained by the maximally mixed state I_d/d.

If

$$\rho = \sum_i \lambda_i |i\rangle\langle i| \tag{1.37}$$

is the spectral decomposition of the state, then

$$\gamma(\rho) = \sum_i \lambda_i^2. \tag{1.38}$$

For a single qubit written in Bloch form, as $d = 2$, Eq. (1.20) gives

$$\gamma(\rho) = \frac{1 + \|\mathbf{r}\|^2}{2}, \qquad \text{with} \qquad \frac{1}{2} \leq \gamma(\rho) \leq 1. \tag{1.39}$$

For the two-qubit polarization system considered throughout this work, $d = 4$, and therefore

$$\frac{1}{4} \leq \gamma(\rho_{AB}) \leq 1. \tag{1.40}$$

Purity quantifies mixedness but does not identify its physical origin. A mixed reduced state may result, for example, from tracing over an entangled subsystem, whereas a mixed global state may arise from statistical averaging or interaction with unresolved degrees of freedom.

1.6.2. Fidelity

Throughout this work, fidelity is defined using the squared Uhlmann convention [13], for two states ρ and σ it is given by

$$F(\rho, \sigma) = \left[\mathrm{Tr} \left(\sqrt{\sqrt{\rho}\, \sigma \, \sqrt{\rho}} \right) \right]^2 . \tag{1.41}$$

When the reference state is pure, $\sigma = |\phi\rangle\langle\phi|$, the expression simplifies to

$$F(\rho, |\phi\rangle) = \langle\phi|\rho|\phi\rangle. \tag{1.42}$$

In this thesis the principal reference is $|\Psi^+\rangle$, so that

$$F_{\Psi^+} = \langle\Psi^+|\rho|\Psi^+\rangle. \tag{1.43}$$

Fidelity measures similarity to a chosen target and must therefore not be interpreted directly as an entanglement measure. A local unitary transformation can substantially change F_{Ψ^+} while leaving the amount of entanglement unchanged.

1.6.3. Concurrence and Tangle

The qualitative distinction between separable and entangled states was introduced in Section 1.2. A quantitative measure is now required to characterize the degree of entanglement of the propagated two-qubit state. Throughout this work, concurrence is adopted as the principal measure of bipartite entanglement [14].

For a pure state

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle, \tag{1.44}$$

the concurrence is

$$C(|\psi\rangle) = 2|ad - bc|. \tag{1.45}$$

It satisfies

$$0 \le C \le 1, \tag{1.46}$$

with $C = 0$ for separable pure states and $C = 1$ for maximally entangled two-qubit states.

For pure bipartite states, concurrence can also be expressed through either reduced density operator,

$$
\begin{aligned}
C(|\psi_{AB}\rangle) &= 2\sqrt{\det \rho_A} = \sqrt{2\left[1 - \mathrm{Tr}(\rho_A^2)\right]} \\
&= 2\sqrt{\det \rho_B} = \sqrt{2\left[1 - \mathrm{Tr}(\rho_B^2)\right]}.
\end{aligned} \tag{1.47}
$$

The equivalence follows from the fact that, for a globally pure bipartite state, ρ_A and ρ_B have the same nonzero eigenvalues.

For a mixed two-qubit state, concurrence is defined through the Wootters spin-flip construction [14]. Defining

$$
\widetilde{\rho} = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y), \tag{1.48}
$$

and denoting by $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4$ the square roots of the eigenvalues of $\rho\widetilde{\rho}$, one obtains

$$
C(\rho) = \max\left(0,\, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\right). \tag{1.49}
$$

The associated tangle is defined as

$$
\tau(\rho) = C(\rho)^2. \tag{1.50}
$$

For the reference Bell state,

$$
C(|\Psi^+\rangle) = 1, \qquad \tau(|\Psi^+\rangle) = 1. \tag{1.51}
$$

Both quantities are invariant under local unitary transformations, consistently with the invariance of bipartite entanglement under local changes of polarization basis.

1.6.4. Entanglement of Formation

The entanglement of formation provides a resource-oriented measure of bipartite entanglement. For two qubits, it can be expressed analytically in terms of the concurrence [14] as

$$E_F(\rho) = h\left(\frac{1+\sqrt{1-C(\rho)^2}}{2}\right), \tag{1.52}$$

where

$$h(x) = -x\log_2 x - (1-x)\log_2(1-x) \tag{1.53}$$

is the binary entropy.

For two-qubit states,

$$0 \le E_F \le 1, \tag{1.54}$$

with $E_F = 0$ for separable states and $E_F = 1$ for maximally entangled Bell states. Because E_F is a monotonic function of concurrence for two-qubit states, the two quantities preserve the same ordering of entanglement, while providing different quantitative representations.

1.7. Complementarity of the State Metrics

The quantities introduced in the previous sections characterize different properties of the same density operator and should therefore not be interpreted interchangeably. Purity quantifies whether a state is pure or mixed but does not measure entanglement, while fidelity quantifies similarity with respect to a selected reference state and can change under coherent local transformations. Concurrence, tangle, and entanglement of formation instead quantify bipartite entanglement.

This distinction is particularly relevant for optical-fiber propagation. A coherent local polarization transformation may alter the correlations measured along fixed laboratory axes and reduce fidelity with respect to the initial Bell state without changing purity or entanglement. By contrast, unresolved statistical fluctuations or coupling between polarization and unobserved degrees of freedom may produce mixedness and a reduction of the distributed entanglement.

The simultaneous evaluation of these quantities therefore provides a state-level description capable of distinguishing coherent transformations from decoherence and entanglement degradation. Their operational significance, however, depends on the application for which the distributed state is intended. For this reason, the following section introduces the BBM92 protocol and the quantities used in this work to connect the characterization of the quantum state with the requirements of entanglement-based quantum communication.

1.8. BBM92 and Operational Requirements

The main operational reference considered in this thesis is the BBM92 entanglement-based quantum key distribution protocol [3]. In its polarization implementation, a source distributes the two photons of an entangled pair to two spatially separated parties, conventionally denoted Alice and Bob. Each party independently selects between alternative polarization measurement bases. Measurements performed using compatible bases provide correlated outcomes that can be used for key generation and for estimating the disturbance introduced along the communication link.

For the polarization encoding adopted throughout this work, the computational basis has already been identified with

$$|0\rangle \equiv |H\rangle, \qquad |1\rangle \equiv |V\rangle,$$

while the complementary diagonal basis is

$$|D\rangle = \frac{|H\rangle + |V\rangle}{\sqrt{2}}, \qquad |A\rangle = \frac{|H\rangle - |V\rangle}{\sqrt{2}}. \tag{1.55}$$

For an ideal entangled state, Alice and Bob obtain the expected correlations in both measurement bases. Propagation through the optical link may modify these correlations and introduce errors. Their magnitude is described by the basis-dependent quantum bit error rates

$$Q_Z, \qquad Q_X, \tag{1.56}$$

associated with measurements in the computational and diagonal bases, respectively. These quantities can be obtained from the joint measurement probabilities of the propagated two-photon state.

The QBER provides the main operational connection between the state distributed through the fiber and the BBM92 protocol. Degradation or misalignment of the expected polarization correlations can increase the measured error rates. If the QBER exceeds the admissible value associated with the adopted security model, the corresponding configuration is no longer considered suitable for secure key distribution.

The objective of this work is therefore not only to quantify the degradation of the distributed entanglement, but also to determine which physical configurations of the optical link remain compatible with the BBM92 operational requirement.

For this purpose, the physical configuration is represented by the parameter vector

$$\boldsymbol{\xi} = \{\xi_1, \xi_2, \ldots, \xi_m\}, \tag{1.57}$$

which collects the relevant source and fiber parameters. Depending on the model, these may include photon spectral bandwidth, spectral correlation, differential group delay, birefringence orientation, and compensation parameters.

For each physical configuration $\boldsymbol{\xi}$, the propagated state is characterized by a corresponding set of quantities

$$\boldsymbol{\lambda}(\boldsymbol{\xi}) = \{\gamma,\, F,\, C,\, Q_Z,\, Q_X,\, \ldots\}, \tag{1.58}$$

where γ, F, and C characterize the state in terms of mixedness, similarity to the reference state, and entanglement, while Q_Z and Q_X provide the direct connection with BBM92 performance.

The relation between the physical link and its resulting performance can therefore be summarized as

$$\boxed{\boldsymbol{\xi} \longrightarrow \boldsymbol{\lambda}(\boldsymbol{\xi})} \tag{1.59}$$

where variations of the physical parameters $\boldsymbol{\xi}$ produce corresponding variations of the state and operational quantities contained in $\boldsymbol{\lambda}$.

A BBM92 operating region can then be defined directly in the physical parameter space by requiring the relevant QBER to remain below an adopted threshold Q_{th},

$$\boxed{\mathcal{R}_{\text{BBM92}} = \{\boldsymbol{\xi} : Q(\boldsymbol{\xi}) < Q_{\text{th}}\}\,.} \tag{1.60}$$

Here, Q_{th} is not an arbitrary state-quality threshold, but is determined by the security assumptions adopted for the BBM92 protocol. When the two measurement bases are considered separately, the same criterion can be applied to the corresponding basis-dependent quantities Q_Z and Q_X.

Concurrence plays a complementary role in this analysis. It provides a compact measure of the surviving bipartite entanglement and is therefore used extensively to characterize the effect of fiber propagation. However, concurrence and QBER are not generally equivalent quantities. In particular, a coherent polarization rotation may change the QBER measured in fixed bases without changing the amount of entanglement.

For this reason, a universal concurrence threshold for BBM92 operation is not assumed. Instead, the numerical analysis investigates how concurrence and QBER vary together for the specific families of states produced by the considered fiber models. When a clear correspondence exists for a particular model, the QBER boundary Q_{th} can then be associated with a corresponding value of concurrence.

The resulting framework has a direct practical interpretation: $\boldsymbol{\xi}$ describes the physical source and fiber configuration, $\boldsymbol{\lambda}$ describes the resulting state and its relevant performance quantities, and $\mathcal{R}_{\mathrm{BBM92}}$ identifies the region of physical parameter space compatible with the adopted BBM92 criterion. This perspective is used in the numerical experiments of Chapter 3 to construct operational maps and investigate how the boundaries of acceptable entanglement distribution depend on the physical parameters of the optical link.

2 | Fiber-Channel Modeling

The mathematical framework introduced in Chapter 1 provides the tools required to describe the evolution and characterization of bipartite polarization states. The purpose of the present chapter is to apply these tools to the progressive construction of models for the propagation of polarization-entangled photon pairs through optical fibers.

The modeling strategy proceeds from analytically controlled effective descriptions toward a more physically accurate frequency-dependent representation of fiber propagation. The first models retain only the polarization degree of freedom and separately describe coherent phase evolution, statistical phase fluctuations, and depolarization. These mechanisms are then combined into an effective polarization channel for the two-arm fiber configuration.

The monochromatic approximation is subsequently relaxed by introducing the spectral structure of the photons and the frequency dependence of birefringent fiber propagation. This leads to a polarization-mode-dispersion (PMD) model in which the complete polarization–frequency evolution remains unitary for an ideal lossless fiber, while an effectively mixed polarization state can emerge after the unresolved spectral degree of freedom is traced out.

Figure 2.1 summarizes the modeling hierarchy adopted throughout the chapter. The different levels represent increasing modeling scope rather than a strict physical derivation from one model to the next.

2.1. Residual Phase Model

A general lossless optical fiber may produce an arbitrary unitary transformation of the polarization state. The first propagation model considered in this work is a reduced description of this general evolution and is motivated by the polarization-compensation procedure employed in the experimental setup.

Polarization compensation is used to realign the dominant output polarization basis with the reference basis of the measurement system. Such compensation does not, however, im-

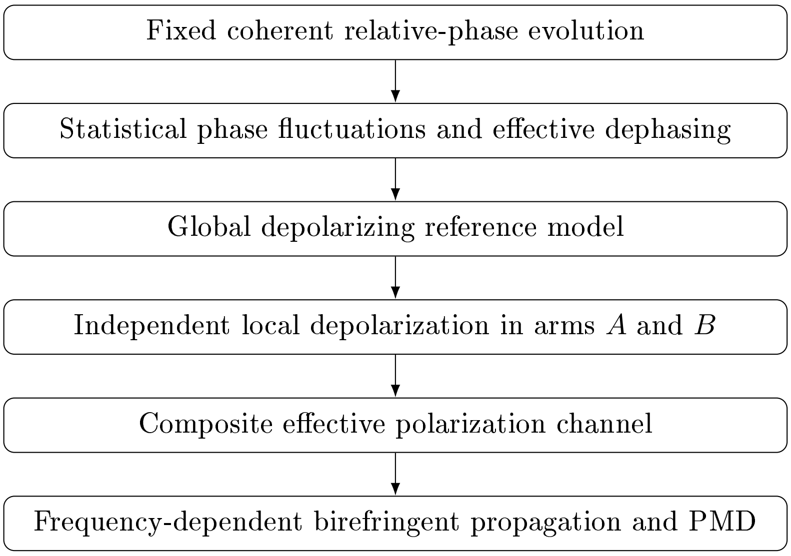


Figure 2.1: Modeling hierarchy adopted in this work. The first levels provide effective polarization-only descriptions of the relevant propagation mechanisms, while the final level explicitly includes the spectral dependence of birefringent fiber propagation.

ply a complete reconstruction or inversion of the full fiber Jones operator[1]. In particular, a residual rotation around the stabilized polarization axis may remain.

A limitation of this compensation procedure arises when the polarization state is monitored primarily through measured populations. Since these measurements provide information on the polarization populations but not directly on their relative phase, a residual phase $e^{i\theta}$ may remain unresolved.

A more complete compensation may be achieved by extending the monitoring system to additional polarization bases, since phase-sensitive measurements can provide information on residual transformations that cannot be inferred from computational-basis populations alone.

Within the computational basis adopted in this thesis, the stabilized polarization axis is identified with the z-axis of the Bloch sphere. The remaining coherent transformation can therefore be represented by a relative phase between the horizontal and vertical polarization components, reducing the general polarization transformation to a single phase parameter.

For one polarization qubit, the corresponding unitary operator is

[1]The Jones matrix is the 2×2 complex matrix describing the transformation of the polarization amplitudes through an optical element or fiber.

$$U(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}, \tag{2.1}$$

defined up to an irrelevant global phase. Equivalently, this transformation corresponds, up to a global phase factor, to a rotation around the z-axis of the Bloch sphere.

At this modeling level, the phase is assumed to be fixed for each propagation realization. Statistical fluctuations of the same residual phase parameter are introduced separately in the following model.

If the two fiber arms introduce residual phases θ_A and θ_B, respectively, their joint action is

$$U_A(\theta_A) \otimes U_B(\theta_B). \tag{2.2}$$

Applying this transformation to the reference Bell state gives

$$|\psi(\theta_A, \theta_B)\rangle = \frac{1}{\sqrt{2}} \left(e^{i\theta_B}|01\rangle + e^{i\theta_A}|10\rangle \right). \tag{2.3}$$

Factoring out the global phase $e^{i\theta_B}$ yields

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + e^{i\theta}|10\rangle \right), \qquad \theta = \theta_A - \theta_B. \tag{2.4}$$

The two independent local phase parameters can therefore be reduced to the single effective parameter θ. This relative phase does not modify the computational-basis populations, but it affects phase-sensitive polarization correlations.

With the convention adopted throughout this work, the same state may be generated by assigning the complete relative transformation to arm A,

$$U_\theta = U_A(\theta) \otimes I_B. \tag{2.5}$$

This convention simplifies the analytical and numerical representation without changing any observable prediction. The parameter θ represents the effective differential residual polarization phase between the two arms.

The term *residual phase model* will be used throughout this work for this one-parameter effective description. Its relation to the more general fixed-frequency birefringent transformation is derived explicitly in Section **??**.

2.1.1. Correlation Structure

For the state in Eq. (2.4), the correlation tensor defined in Chapter 1 is

$$T(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{2.6}$$

The phase therefore rotates the transverse correlation structure in the xy-plane while leaving the longitudinal anticorrelation unchanged:

$$T_{xx} = T_{yy} = \cos\theta, \qquad T_{xy} = -\sin\theta, \qquad T_{yx} = \sin\theta, \qquad T_{zz} = -1. \tag{2.7}$$

The transformation is local and unitary. Consequently,

$$\gamma = 1, \qquad C = 1, \qquad \tau = 1, \qquad E_F = 1 \tag{2.8}$$

for every value of θ.

The fidelity with respect to the original Bell state is instead

$$F_{\Psi^+} = \frac{1+\cos\theta}{2}. \tag{2.9}$$

Thus, a reduction in fixed-reference Bell-state fidelity does not by itself imply entanglement degradation. It may result entirely from a coherent change in the orientation of the state relative to the selected reference basis.

The residual phase model therefore establishes the first important reference case of this thesis: coherent changes in fixed-basis measurements must be distinguished from genuine decoherence and entanglement degradation.

2.2. Statistical Phase Fluctuations and Dephasing

The fixed-phase description assumes that the relative phase is known and remains constant over the relevant observation interval. In practice, birefringence and optical-path conditions may vary between successive photon-pair realizations. If these fluctuations are not resolved by the measurement system, the observed state must be obtained by averaging over the corresponding phase realizations.

Let p_k denote the probability associated with the phase θ_k. The ensemble density operator is

$$\rho_{AB}^{\text{ens}} = \sum_k p_k |\psi(\theta_k)\rangle\langle\psi(\theta_k)|, \qquad \sum_k p_k = 1. \tag{2.10}$$

The phase statistics can be summarized through the complex coherence parameter

$$\mu = \left\langle e^{i\theta} \right\rangle = \sum_k p_k e^{i\theta_k} = \langle \cos\theta \rangle + i\langle \sin\theta \rangle. \tag{2.11}$$

For a continuous probability density $P(\theta)$,

$$\mu = \int d\theta \, P(\theta) e^{i\theta}. \tag{2.12}$$

It is useful to write

$$\mu = |\mu| e^{i\phi_\mu}. \tag{2.13}$$

The magnitude $|\mu|$ quantifies the coherence surviving the statistical averaging, while the argument ϕ_μ represents the residual coherent phase component of the state.

In the computational basis, the resulting state is

$$\rho_{AB}^{\text{ens}} = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & \mu^* & 0 \\ 0 & \mu & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{2.14}$$

The diagonal populations remain unchanged, whereas the off-diagonal coherence terms are multiplied by μ. The model therefore corresponds to a random-unitary dephasing process together with a possible residual coherent phase rotation when $\arg(\mu) \neq 0$.

2.2.1. Correlations and State Metrics

The correlation tensor becomes

$$T_\mu = \begin{pmatrix} \mathrm{Re}(\mu) & -\mathrm{Im}(\mu) & 0 \\ \mathrm{Im}(\mu) & \mathrm{Re}(\mu) & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{2.15}$$

The magnitude $|\mu|$ therefore determines the surviving transverse correlation amplitude, whereas the argument of μ determines its remaining orientation.

For this family of states,

$$\gamma = \frac{1+|\mu|^2}{2}, \tag{2.16}$$

$$F_{\Psi^+} = \frac{1+\mathrm{Re}(\mu)}{2}, \tag{2.17}$$

and

$$C = |\mu|, \qquad \tau = |\mu|^2. \tag{2.18}$$

The corresponding entanglement of formation is

$$E_F = h\left(\frac{1+\sqrt{1-|\mu|^2}}{2}\right), \tag{2.19}$$

where $h(x)$ is the binary entropy introduced in Chapter 1.

The limiting cases provide a direct physical interpretation. For a deterministic phase, $|\mu| = 1$, and the state remains pure and maximally entangled.

For a phase uniformly distributed over $[0, 2\pi]$, $\mu = 0$, and the transverse coherences vanish completely. The resulting state is

$$\rho_{\mu=0} = \frac{1}{2}\left(|01\rangle\langle 01| + |10\rangle\langle 10|\right), \tag{2.20}$$

which retains perfect classical anticorrelation in the computational basis but is no longer entangled.

Intermediate phase distributions produce partially coherent mixed states.

This model represents decoherence at the reduced polarization level, but it does not yet specify the microscopic physical mechanism responsible for the unresolved phase distribution. A frequency-resolved mechanism producing related losses of polarization coherence is introduced later in this chapter.

2.3. Depolarizing Reference Models

Statistical phase averaging suppresses specific coherence terms while preserving the longitudinal population correlation. A complementary analytical reference is provided by isotropic depolarization, which suppresses polarization information uniformly in all directions.

The depolarizing models introduced here are phenomenological. They do not represent a microscopic model of PMD. Their purpose is to provide analytically controlled descriptions of polarization degradation before the physical origin of decoherence is treated explicitly through frequency-dependent propagation.

2.3.1. Global Depolarizing Baseline

For a two-qubit density operator, the global depolarizing channel is defined as

$$\mathcal{D}_p^4(\rho) = (1-p)\rho + p\frac{I_4}{4}, \qquad 0 \le p \le 1. \tag{2.21}$$

For a Bell-state input,

$$\rho(p) = (1-p)|\Psi^+\rangle\langle\Psi^+| + p\frac{I_4}{4}. \tag{2.22}$$

The corresponding purity is

$$\gamma = 1 - \frac{3}{2}p + \frac{3}{4}p^2, \tag{2.23}$$

the Bell-state fidelity is

$$F_{\Psi^+} = 1 - \frac{3}{4}p, \tag{2.24}$$

and the concurrence is

$$C = \max\left(0,\, 1 - \frac{3}{2}p\right). \tag{2.25}$$

All bipartite Pauli correlations are contracted according to

$$T_{\text{out}} = (1-p)T_{\text{in}}. \tag{2.26}$$

The global channel is useful as a validation benchmark, but it is not directly aligned with the geometry of the physical system considered in this thesis, where the two photons propagate through separate fibers.

2.3.2. Local Two-Arm Depolarization

A description more closely aligned with the two-arm geometry description assigns an independent depolarizing channel to each propagation arm.

The parameter convention adopted throughout this work is defined directly by the Bloch-vector contraction. A single-qubit depolarizing channel is written as

$$\mathcal{D}_p(\rho) = (1-p)\rho + p\frac{I_2}{2}, \tag{2.27}$$

so that $p = 0$ corresponds to no depolarization and $p = 1$ to complete depolarization.

Its linear extension to a generic operator X is

$$\mathcal{D}_p(X) = (1-p)X + p\frac{I_2}{2}\mathrm{Tr}(X). \tag{2.28}$$

For a single-qubit state written in Bloch form,

$$\rho = \frac{1}{2}\left(I + \mathbf{r}\cdot\boldsymbol{\sigma}\right), \tag{2.29}$$

the channel gives

$$\mathcal{D}_p(\rho) = \frac{1}{2}\left[I + (1-p)\mathbf{r}\cdot\boldsymbol{\sigma}\right]. \tag{2.30}$$

The Bloch vector therefore undergoes the isotropic contraction

$$\mathbf{r} \longmapsto (1-p)\mathbf{r}. \tag{2.31}$$

For two independent fiber arms,

$$\rho_{AB}^{\text{out}} = (\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left(\rho_{AB}^{\text{in}}\right). \tag{2.32}$$

Although the effective noise acts locally, the channel acts on the complete bipartite density operator. The entanglement and two-photon correlations are properties of the joint state and cannot be reconstructed from the two reduced states alone.

2.3.3. Correlation-Survival Factor

The single-qubit depolarizing channel acts on the Pauli basis as

$$\mathcal{D}_p(I) = I, \qquad \mathcal{D}_p(\sigma_i) = (1-p)\sigma_i, \qquad i \in \{x, y, z\}. \tag{2.33}$$

Consequently, each two-body Pauli correlation is multiplied by the factor

$$\eta = (1-p_A)(1-p_B). \tag{2.34}$$

The correlation tensor therefore satisfies

$$T_{\text{out}} = \eta T_{\text{in}}, \tag{2.35}$$

and, for arbitrary local measurement axes,

$$E_{\text{out}}(\mathbf{a}, \mathbf{b}) = \eta E_{\text{in}}(\mathbf{a}, \mathbf{b}). \tag{2.36}$$

The factor η will be used throughout the remainder of the effective fiber-channel models as the survival factor of bipartite polarization correlations.

2.4. Composite Effective Fiber Channel

The fixed relative-phase model and the local depolarizing model represent two different classes of propagation effects. The former is coherent and rotates the correlation structure, whereas the latter is non-unitary at the polarization level and suppresses its magnitude.

They are combined through the effective channel

$$\rho_{AB}^{\mathrm{out}} = (\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left[U_\theta \rho_{AB}^{\mathrm{in}} U_\theta^\dagger \right]. \tag{2.37}$$

For isotropic depolarization, the local depolarizing channel is covariant under single-qubit unitary transformations. Consequently, for the phase transformation considered here,

$$(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left[U_\theta \rho U_\theta^\dagger \right] = U_\theta \left[(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) (\rho) \right] U_\theta^\dagger. \tag{2.38}$$

Thus, within this isotropic model, the order of phase evolution and depolarization does not affect the final state. This property is specific to the adopted channel and does not hold for arbitrary anisotropic noise mechanisms.

2.4.1. Fixed-Phase Composite Model

For the maximally entangled input family considered here, the reduced states of both subsystems remain $I_2/2$. The action of the two local depolarizing channels can therefore be written compactly as

$$\rho_{AB}^{\mathrm{out}} = \eta |\psi(\theta)\rangle\langle\psi(\theta)| + (1-\eta)\frac{I_4}{4}. \tag{2.39}$$

The corresponding correlation tensor is

$$T_{\mathrm{out}} = \eta \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{2.40}$$

The parameter θ therefore controls the orientation of the transverse correlations, whereas η controls their magnitude.

The corresponding purity is

$$\gamma = \frac{1+3\eta^2}{4}, \tag{2.41}$$

while the Bell-state fidelity is

$$F_{\Psi^+} = \frac{1-\eta}{4} + \eta\frac{1+\cos\theta}{2}. \tag{2.42}$$

The concurrence is

$$C = \max\left(0, \frac{3\eta-1}{2}\right). \tag{2.43}$$

2.4.2. Statistical Composite Model

The composite description can be extended by allowing the relative phase to fluctuate between realizations. The ensemble-averaged state is

$$\rho_{AB}^{\text{ens}} = \sum_k p_k \left(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}\right) \left[U_{\theta_k} \rho_{AB}^{\text{in}} U_{\theta_k}^{\dagger}\right]. \tag{2.44}$$

Using the covariance property in Eq. (2.38),

$$\rho_{AB}^{\text{ens}} = \left(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}\right) \left[\sum_k p_k U_{\theta_k} \rho_{AB}^{\text{in}} U_{\theta_k}^{\dagger}\right]. \tag{2.45}$$

For the random-phase family of Section 2.2, this reduces to

$$\rho_{AB}^{\text{ens}} = \eta \rho_{AB}^{\mu} + (1-\eta)\frac{I_4}{4}, \tag{2.46}$$

where ρ_{AB}^{μ} denotes the state in Eq. (2.14).

The output correlation tensor is

$$T_{\text{out}}^{\text{ens}} = \eta \begin{pmatrix} \mathrm{Re}(\mu) & -\mathrm{Im}(\mu) & 0 \\ \mathrm{Im}(\mu) & \mathrm{Re}(\mu) & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{2.47}$$

The two mechanisms therefore leave distinguishable signatures. Phase averaging suppresses the transverse coherence through μ, whereas local isotropic depolarization multiplies all bipartite correlations by η.

The purity becomes

$$\gamma = \frac{1 + \eta^2 \left(1 + 2|\mu|^2\right)}{4}, \tag{2.48}$$

the fidelity is

$$F_{\Psi^+} = \frac{1-\eta}{4} + \eta \frac{1 + \mathrm{Re}(\mu)}{2}, \tag{2.49}$$

and the concurrence is

$$C = \max\left(0,\, \eta|\mu| - \frac{1-\eta}{2}\right). \tag{2.50}$$

At the level of transverse polarization correlations, the progression of the effective models can therefore be summarized as

$$\boxed{e^{i\theta} \quad \longrightarrow \quad \mu = \left\langle e^{i\theta} \right\rangle \quad \longrightarrow \quad \eta\mu} \tag{2.51}$$

corresponding respectively to coherent phase evolution, statistical dephasing, and the additional suppression produced by isotropic depolarization. The effective model is analytically transparent and useful for distinguishing coherent phase evolution, statistical dephasing, and isotropic polarization degradation. Its main limitation is that the parameters μ, p_A, and p_B describe phenomenological behavior rather than being derived directly from the spectral and birefringent properties of the physical fibers.

2.5. From Effective Channels to Frequency-Dependent Propagation

The effective models introduced above treat each photon exclusively as a polarization qubit and assume that the relevant fiber action can be represented without explicitly retaining optical frequency.

This approximation is appropriate when the polarization transformation varies negligibly across the photon bandwidth, or when the unresolved physical mechanisms can be represented adequately by phenomenological channel parameters.

The effective models nevertheless omit several properties of real birefringent fiber propagation. In particular, they contain no explicit optical bandwidth, principal states of polarization, differential group delay, or frequency-dependent polarization transformation. Moreover, isotropic depolarization cannot reproduce the directional character of PMD: a state aligned with a principal state may remain fully polarized, whereas a superposition of the principal states may become correlated with arrival time and lose reduced polarization coherence.

The polarization-only models also do not distinguish between two physically different origins of mixedness:

- averaging over different propagation realizations occurring at different times;
- polarization becoming correlated with unresolved spectral or temporal degrees of freedom within an individual photon wave packet.

Both mechanisms can produce mixed reduced polarization states, but their physical interpretations are different.

SPDC photons have finite optical bandwidth. Their spectral structure is determined by the pump envelope and by the phase-matching properties of the nonlinear source [15]. Different frequency components may therefore experience different birefringent transformations during propagation.

The local polarization operator must consequently be generalized to

$$U_K \longrightarrow U_K(\omega_K), \qquad K \in \{A, B\}. \tag{2.52}$$

For a resolved pair of frequencies,

$$U_{AB}(\omega_A, \omega_B) = U_A(\omega_A) \otimes U_B(\omega_B). \tag{2.53}$$

The complete polarization–frequency evolution of an ideal lossless fiber remains unitary. Nevertheless, frequency-dependent propagation can correlate polarization with frequency and arrival time. If these degrees of freedom are not resolved by the measurement system, the observable polarization state can become mixed.

This provides the physical connection with the phenomenological models developed above: an effective non-unitary polarization channel can emerge from an underlying unitary evolution when only part of the complete optical system is observed.

2.5.1. Illustrative Effect of Frequency-Dependent Birefringence

The physical consequence of the frequency dependence introduced in Eq. (2.52) can be understood qualitatively by considering different spectral components of the same photon.

In a birefringent medium, the polarization transformation generally depends on optical frequency. Consequently, two frequency components ω_1 and ω_2 need not undergo the same transformation,

$$U(\omega_1) \neq U(\omega_2). \tag{2.54}$$

For example, if the input polarization contains components along both local birefringent eigenmodes, the two components accumulate a relative phase during propagation. Because the birefringence is generally dispersive, this relative phase depends on frequency. Different parts of the photon spectrum can therefore emerge from the fiber with different polarization states.

For a sufficiently narrow optical spectrum, the transformation may be considered approximately constant over the occupied bandwidth,

$$U(\omega) \simeq U(\omega_0), \tag{2.55}$$

where ω_0 denotes the central optical frequency. In this limit, all spectral components undergo essentially the same polarization transformation and the polarization dynamics can be described accurately by a single-frequency unitary operator.

When the frequency dependence cannot be neglected, however, different spectral components experience different polarization transformations. Polarization then becomes correlated with the spectral degree of freedom and, equivalently in the time domain, with the temporal structure of the photon wave packet.

The complete optical evolution can still be unitary and preserve the purity of the full state. Nevertheless, a detector that does not resolve the spectral or temporal degree of freedom has access only to the polarization subsystem. The corresponding reduced polarization state may therefore exhibit a loss of coherence and become mixed.

This mechanism is fundamentally different from the phenomenological isotropic depolarizing channel introduced in Section 2.3. Here, the apparent polarization decoherence originates from coherent frequency-dependent propagation followed by the neglect of an unresolved degree of freedom.

The spectral Hilbert space, the finite-bandwidth two-photon state, and the corresponding reduction to an effective polarization state are introduced formally in the following section. The relation between frequency-dependent birefringence, differential phase, differential group delay, and PMD is then developed subsequently.

2.6. Spectral Description of SPDC Photon Pairs

The polarization-only description introduced previously treats each photon as a two-level system,

$$\mathcal{H}_{\mathrm{pol},K} \simeq \mathbb{C}^2, \qquad K \in \{A, B\}. \tag{2.56}$$

This description is sufficient as long as the optical transformation can be regarded as independent of frequency. Polarization-mode dispersion, however, originates precisely from a frequency-dependent polarization transformation. The spectral degree of freedom must therefore be included explicitly in the state space [5, 7, 10].

For each photon K, the Hilbert space is consequently extended to

$$\mathcal{H}_K = \mathcal{H}_{\omega_K} \otimes \mathcal{H}_{\mathrm{pol},K}, \qquad K \in \{A, B\}, \tag{2.57}$$

where $\mathcal{H}_{\omega_K}$ describes the continuous spectral degree of freedom and $\mathcal{H}_{\mathrm{pol},K}$ describes the polarization qubit. The complete two-photon space is therefore

$$\mathcal{H}_{\mathrm{tot}} = \mathcal{H}_A \otimes \mathcal{H}_B = \left(\mathcal{H}_{\omega_A} \otimes \mathcal{H}_{\mathrm{pol},A}\right) \otimes \left(\mathcal{H}_{\omega_B} \otimes \mathcal{H}_{\mathrm{pol},B}\right). \tag{2.58}$$

For convenience, the tensor factors are reordered throughout this work so that the two spectral degrees of freedom and the two polarization degrees of freedom are grouped together:

$$\mathcal{H}_{\mathrm{tot}} = \underbrace{\left(\mathcal{H}_{\omega_A} \otimes \mathcal{H}_{\omega_B}\right)}_{\mathcal{H}_{\omega,\mathrm{tot}}} \otimes \underbrace{\left(\mathcal{H}_{\mathrm{pol},A} \otimes \mathcal{H}_{\mathrm{pol},B}\right)}_{\mathcal{H}_{\mathrm{pol,tot}}}. \tag{2.59}$$

The two orderings are physically equivalent; the second one is adopted because it makes the separation between spectral and polarization degrees of freedom explicit and allows the spectral partial trace introduced later to be written naturally.

Frequency eigenstates are denoted by $|\omega_K\rangle$. Since frequency is a continuous variable, these states obey Dirac-delta normalization,

$$\langle\omega_K|\omega'_K\rangle = \delta(\omega_K - \omega'_K), \qquad K \in \{A, B\}, \tag{2.60}$$

together with the completeness relation

$$\int d\omega_K\, |\omega_K\rangle\langle\omega_K| = I_{\omega_K}. \tag{2.61}$$

The corresponding two-photon frequency basis states are

$$|\omega_A, \omega_B\rangle = |\omega_A\rangle \otimes |\omega_B\rangle, \tag{2.62}$$

and satisfy

$$\langle\omega_A, \omega_B|\omega'_A, \omega'_B\rangle = \delta(\omega_A - \omega'_A)\delta(\omega_B - \omega'_B). \tag{2.63}$$

These relations are the continuous-variable counterpart of the orthogonality and completeness relations used for discrete quantum bases.

The finite-bandwidth spectral state generated by the SPDC source can then be expanded in this basis as

$$|F\rangle_{\omega_A\omega_B} = \iint d\omega_A\, d\omega_B\, F(\omega_A, \omega_B)|\omega_A, \omega_B\rangle, \tag{2.64}$$

where

$$F(\omega_A, \omega_B) = \langle\omega_A, \omega_B|F\rangle \tag{2.65}$$

is the joint spectral amplitude (JSA) of the photon pair. It therefore plays the role of a two-photon wavefunction in the frequency representation: for each pair (ω_A, ω_B), it specifies the complex probability amplitude associated with generating photon A at frequency ω_A and photon B at frequency ω_B [15].

The spectral state is normalized according to

$$\iint d\omega_A\, d\omega_B\, |F(\omega_A, \omega_B)|^2 = 1. \tag{2.66}$$

The corresponding joint spectral intensity (JSI) is

$$J(\omega_A, \omega_B) = |F(\omega_A, \omega_B)|^2, \tag{2.67}$$

which represents the joint probability density of the two photon frequencies. In particular,

$$dP = J(\omega_A, \omega_B)\, d\omega_A\, d\omega_B \tag{2.68}$$

is the probability of finding the two photons within infinitesimal spectral intervals around ω_A and ω_B, respectively.

The corresponding marginal spectra are obtained by integrating over the frequency of the other photon,

$$P_A(\omega_A) = \int d\omega_B\, J(\omega_A, \omega_B), \tag{2.69}$$

$$P_B(\omega_B) = \int d\omega_A\, J(\omega_A, \omega_B). \tag{2.70}$$

The joint spectral structure originates from the physical SPDC generation process. In spontaneous parametric down-conversion, one pump photon is converted into two lower-frequency photons conventionally referred to as signal and idler photons. In the two-arm notation adopted here, these photons are associated with subsystems A and B [15].

Energy conservation imposes

$$\omega_p = \omega_A + \omega_B. \tag{2.71}$$

For an ideal monochromatic pump of angular frequency $\omega_{p,0}$, this constraint fixes the generated frequencies to the line

$$\omega_A + \omega_B = \omega_{p,0}. \tag{2.72}$$

A realistic pump instead possesses a finite spectral bandwidth. Its spectral amplitude is described by the pump envelope $\alpha_p(\omega_p)$, which, after imposing energy conservation, enters the two-photon state as

$$\alpha_p(\omega_A + \omega_B). \tag{2.73}$$

The generated frequencies must also satisfy the phase-matching condition of the nonlinear medium. Introducing the longitudinal wave-vector mismatch

$$\Delta k(\omega_A, \omega_B) = k_p(\omega_A + \omega_B) - k_A(\omega_A) - k_B(\omega_B), \tag{2.74}$$

a commonly used form for a uniform nonlinear interaction of length L is

$$\Phi(\omega_A, \omega_B) = \operatorname{sinc}\left[\frac{\Delta k(\omega_A, \omega_B)L}{2}\right] \exp\left[i\frac{\Delta k(\omega_A, \omega_B)L}{2}\right], \tag{2.75}$$

up to convention-dependent phase factors and possible additional terms associated with the specific nonlinear-crystal configuration [15].

The joint spectral amplitude can therefore be written schematically as

$$F(\omega_A, \omega_B) = \mathcal{N}\, \alpha_p(\omega_A + \omega_B)\Phi(\omega_A, \omega_B), \tag{2.76}$$

where $\mathcal{N}$ is chosen such that Eq. (2.66) is satisfied [15, 16].

Equation (2.76) separates two physically different constraints. The pump envelope determines which values of the sum frequency $\omega_A + \omega_B$ are supplied by the pump, whereas $\Phi(\omega_A, \omega_B)$ determines which of these frequency pairs are efficiently generated by the phase-matching properties of the nonlinear medium. The observed joint spectrum results from the overlap of these two spectral constraints.

Consequently, the generated frequencies need not be statistically independent. For example, a narrowband pump strongly constrains $\omega_A + \omega_B$, so that an increase in one photon frequency must be accompanied by a corresponding decrease in the other. This produces the frequency anticorrelation commonly encountered in SPDC photon pairs [15].

More generally, complete spectral separability requires the joint spectral amplitude to factorize as

$$F(\omega_A, \omega_B) = f_A(\omega_A) f_B(\omega_B). \tag{2.77}$$

When such a decomposition is not possible, the spectral state cannot be written as a product of two independent single-photon spectral states. This distinction is important because the phase of the JSA, and not only its intensity $J = |F|^2$, can contribute to spectral correlations.

2.6.1. Input Polarization–Frequency State

The propagation model developed in this work introduces an additional assumption concerning the relation between the spectral and polarization degrees of freedom at the source output. These degrees of freedom are assumed to be initially separable, so that the polarization state is the same for every generated frequency pair.

In particular, the two polarization alternatives forming the Bell state are assumed to possess the same joint spectral amplitude. Under this condition, the multimode two-photon state can be factorized into independent spectral and polarization components [17].

Physically, this assumption corresponds to neglecting source-induced polarization–frequency distinguishability, or equivalently to assuming that such distinguishability has been sufficiently suppressed through source design, compensation, and spectral filtering. In practical polarization-entangled SPDC sources, spectral differences between alternative generation processes can otherwise provide distinguishing information and reduce the observed polarization entanglement [18].

A factorized temporal–polarization input state is also commonly adopted in theoretical treatments of PMD, where coupling between these degrees of freedom is subsequently generated by propagation [7].

For the reference Bell state,

$$|\Psi^{\text{in}}_{\text{tot}}\rangle = |F\rangle_{\omega_A\omega_B} \otimes |\Psi^+\rangle_{\text{pol}}. \tag{2.78}$$

Equivalently, using the spectral expansion introduced in Eq. (2.64),

$$|\Psi^{\text{in}}_{\text{tot}}\rangle = \iint d\omega_A\, d\omega_B\, F(\omega_A, \omega_B)|\omega_A, \omega_B\rangle \otimes |\Psi^+\rangle_{\text{pol}}. \tag{2.79}$$

More generally, allowing the polarization state itself to be mixed,

$$\rho^{\text{in}}_{\text{tot}} = |F\rangle\langle F| \otimes \rho^{\text{pol,in}}_{AB}. \tag{2.80}$$

For the reference pure Bell-state input,

$$\rho_{AB}^{\mathrm{pol,in}} = |\Psi^+\rangle\langle\Psi^+|. \tag{2.81}$$

This factorization is an explicit modeling assumption rather than a general property of type-II SPDC sources. It states that the two polarization alternatives forming the Bell state share the same joint spectral state, so that the source initially carries no spectral information capable of distinguishing them.

Initial polarization–frequency correlations are therefore neglected. Within the present model, any such coupling appearing after transmission can consequently be attributed to the frequency dependence of the fiber propagation [5, 7].

This factorized state provides the starting point for the PMD model developed in the following sections. Once the local fiber transformations become frequency-dependent, different frequency pairs generally undergo different polarization transformations, and the factorized structure of the input state need not be preserved.

2.6.2. Frequency-Controlled Fiber Evolution

An ideal stationary lossless fiber preserves optical frequency while applying a frequency-dependent polarization transformation. In the tensor ordering of Eq. (2.59), the complete two-arm propagation operator can therefore be written directly as

$$\mathcal{U}_{AB} = \iint d\omega_A \, d\omega_B \, |\omega_A, \omega_B\rangle\langle\omega_A, \omega_B| \otimes \left[U_A(\omega_A) \otimes U_B(\omega_B)\right]. \tag{2.82}$$

The operator in Eq. (2.82) is diagonal in the frequency basis. The projector $|\omega_A, \omega_B\rangle\langle\omega_A, \omega_B|$ does not represent a physical measurement of frequency. Rather, it ensures that each spectral component of the state is associated with the polarization transformation corresponding to that particular frequency pair. The frequencies themselves are left unchanged, while the polarization transformation depends on their values.

This action can be made explicit by considering first a state with a definite frequency pair (ω'_A, ω'_B) and an arbitrary two-photon polarization state $|\chi\rangle$,

$$|\Psi\rangle = |\omega'_A, \omega'_B\rangle \otimes |\chi\rangle. \tag{2.83}$$

Applying Eq. (2.82) gives

$$\mathcal{U}_{AB}|\Psi\rangle = \iint d\omega_A\, d\omega_B\, |\omega_A, \omega_B\rangle\langle\omega_A, \omega_B|\omega'_A, \omega'_B\rangle \otimes [U_A(\omega_A) \otimes U_B(\omega_B)]\, |\chi\rangle. \tag{2.84}$$

Using the orthogonality relation

$$\langle\omega_A, \omega_B|\omega'_A, \omega'_B\rangle = \delta(\omega_A - \omega'_A)\delta(\omega_B - \omega'_B), \tag{2.85}$$

the two integrals collapse, yielding

$$\mathcal{U}_{AB}\left(|\omega'_A, \omega'_B\rangle \otimes |\chi\rangle\right) = |\omega'_A, \omega'_B\rangle \otimes [U_A(\omega'_A) \otimes U_B(\omega'_B)]\, |\chi\rangle. \tag{2.86}$$

Equation (2.86) provides the direct physical interpretation of the full propagation operator. The fiber does not convert the photons from one optical frequency to another. Instead, the frequency pair determines which local polarization transformations are experienced by the two photons. In this sense, Eq. (2.82) can be interpreted as a continuous frequency-controlled polarization transformation.

For a finite-bandwidth input state, many frequency pairs are present simultaneously. Each spectral component therefore undergoes, in general, a different polarization transformation. This is the mechanism through which frequency-dependent propagation can establish correlations between spectral and polarization degrees of freedom.

After propagation,

$$\rho_{\mathrm{tot}}^{\mathrm{out}} = \mathcal{U}_{AB}\rho_{\mathrm{tot}}^{\mathrm{in}}\mathcal{U}_{AB}^{\dagger}. \tag{2.87}$$

If the spectral degrees of freedom are not resolved, the observable polarization state is obtained by tracing them out:

$$\rho_{AB}^{\mathrm{pol,out}} = \mathrm{Tr}_{\omega_A,\omega_B}\left[\rho_{\mathrm{tot}}^{\mathrm{out}}\right]. \tag{2.88}$$

Using the orthogonality of the frequency eigenstates and performing the partial trace over the unresolved spectral degrees of freedom gives [5, 9]

$$\rho_{AB}^{\mathrm{pol,out}} = \iint d\omega_A\, d\omega_B\, J(\omega_A, \omega_B)\, [U_A(\omega_A) \otimes U_B(\omega_B)]\, \rho_{AB}^{\mathrm{pol,in}}\, [U_A(\omega_A) \otimes U_B(\omega_B)]^{\dagger}. \tag{2.89}$$

Equation (2.89) is the central connection between the frequency-dependent physical propagation model and the effective polarization channel.

Only the joint spectral intensity $J = |F|^2$ appears in this expression. Under the assumptions made above, the spectral phase of the JSA cancels after the frequency-preserving propagation and complete spectral trace. This simplification relies on the initial polarization–frequency factorization and on the absence of frequency conversion in the fiber.

The result can be interpreted as a continuous random-unitary channel on the polarization subsystem: each frequency pair produces a local unitary transformation, while the unresolved spectral distribution determines the statistical weighting [5, 10].

2.6.3. Nature of the ResidualPolarization Decoherence

The reduced evolution in Eq. (2.89) should not in general be identified with a standard isotropic depolarizing channel. Its specific form depends on the frequency dependence of the local polarization transformations.

If the birefringence eigenaxes remain fixed while only the differential phase varies with frequency, tracing over the unresolved spectrum produces a dephasing-type channel in that eigenpolarization basis: the corresponding populations are preserved while the off-diagonal coherence terms are reduced.

More general frequency-dependent birefringent propagation may also involve a variation of the effective polarization rotation axes across the spectrum. In this case, the spectral average can contract the polarization state along different Bloch-sphere directions and produce a more general anisotropic polarization decoherence.

Such a reduction may be described as depolarization in the optical sense, since the degree of polarization and the reduced-state purity can decrease, but it does not necessarily coincide with the quantum depolarizing channel

$$\mathcal{D}_p(\rho) = (1-p)\rho + p\frac{I}{2}, \tag{2.90}$$

which represents an isotropic contraction toward the maximally mixed state. Only under additional symmetry or statistical assumptions can the frequency-averaged propagation reduce to such an isotropic model.

The PMD description developed below should therefore be interpreted more generally as frequency-induced polarization decoherence arising from an underlying unitary polarization–frequency evolution [5, 9, 19].

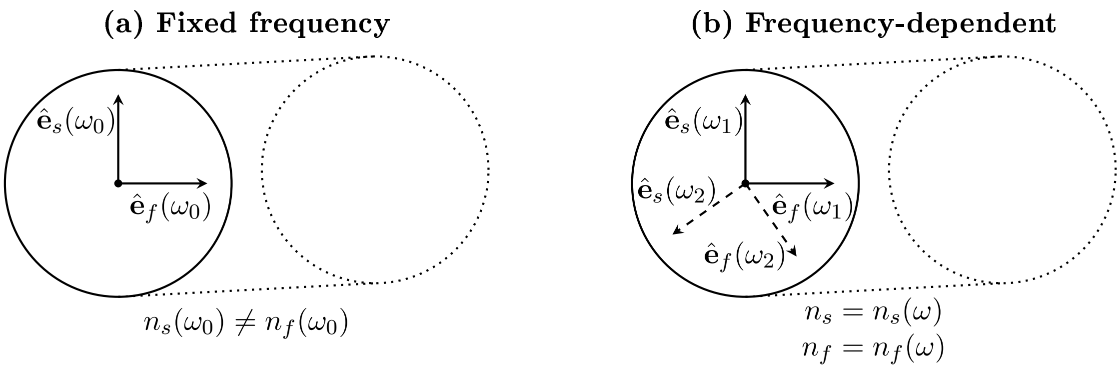


Figure 2.2: Schematic representation of local fiber birefringence at a fixed frequency and over a finite spectral bandwidth. At ω_0, the fiber supports two orthogonal slow and fast polarization eigenmodes characterized by different effective refractive indices. When the frequency dependence of birefringence is taken into account, different spectral components may experience different birefringent parameters; in panel (b), the solid and dashed orthogonal pairs schematically represent the eigenmodes associated with ω_1 and ω_2, respectively.

2.7. Birefringence, Differential Phase, and Group Delay

Polarization-mode dispersion originates from the frequency dependence of birefringence in optical fibers. Even a nominally single-mode fiber supports two orthogonal polarization modes whose propagation constants are generally different [7, 8, 19].

Figure 2.2 schematically illustrates the distinction between a fixed-frequency description of birefringence and the more general frequency-dependent picture. At a given frequency, the fiber is characterized by two orthogonal polarization eigenmodes with different effective refractive indices. Over a finite spectral bandwidth, these birefringent properties may vary with frequency, leading to frequency-dependent polarization evolution.

Consider a locally uniform birefringent segment with slow and fast polarization eigenmodes. Their propagation constants are denoted by $\beta_s(\omega)$ and $\beta_f(\omega)$, respectively. The differential propagation constant is

$$\Delta\beta(\omega) = \beta_s(\omega) - \beta_f(\omega). \tag{2.91}$$

Introducing the corresponding effective refractive indices,

$$\beta_{s,f}(\omega) = \frac{\omega}{c} n_{s,f}(\omega), \tag{2.92}$$

one obtains

$$\Delta\beta(\omega) = \frac{\omega}{c}\Delta n_{\text{eff}}(\omega), \qquad \Delta n_{\text{eff}} = n_s - n_f. \tag{2.93}$$

For a uniform segment of length L_j, the accumulated differential phase is

$$\Delta\phi_j(\omega) = L_j \Delta\beta_j(\omega). \tag{2.94}$$

The corresponding beat length is

$$L_{b,j} = \frac{2\pi}{|\Delta\beta_j|}, \tag{2.95}$$

representing the propagation distance required to accumulate a differential phase of 2π.

Because the differential phase depends on frequency, the two eigenmodes also possess different group delays. The signed group-delay difference of the segment is

$$\Delta\tau_j = \left.\frac{d\Delta\phi_j}{d\omega}\right|_{\omega_0} = L_j \left.\frac{d\Delta\beta_j}{d\omega}\right|_{\omega_0}. \tag{2.96}$$

Using Eq. (2.93),

$$\Delta\tau_j = \frac{L_j}{c}\left[\Delta n_{\text{eff}}(\omega_0) + \omega_0 \left.\frac{d\Delta n_{\text{eff}}}{d\omega}\right|_{\omega_0}\right]. \tag{2.97}$$

The magnitude $|\Delta\tau_j|$ is the differential group delay of the segment. This establishes the physical chain connecting local birefringence to the frequency-dependent polarization evolution:

$$\Delta n_{\text{eff}} \longrightarrow \Delta\beta \longrightarrow \Delta\phi \longrightarrow \frac{d\Delta\phi}{d\omega} \longrightarrow \Delta\tau.$$

2.8. Single Frequency-Dependent Birefringent Segment

The relations introduced in the previous section describe how local birefringence produces a differential phase and, through its frequency dependence, a differential group delay. These quantities can now be incorporated into an operator description of the polarization evolution.

Consider a locally uniform, linear, and lossless birefringent fiber segment of length L_j. Its polarization evolution can be represented by a frequency-dependent unitary Jones operator [20]. With the phase convention adopted in this work,

$$U_j(\omega) = e^{i\bar{\phi}_j(\omega)} \exp\left[\frac{i}{2}\Delta\phi_j(\omega)\hat{\mathbf{n}}_j \cdot \boldsymbol{\sigma}\right], \tag{2.98}$$

where $\bar{\phi}_j(\omega)$ is a polarization-independent phase, $\Delta\phi_j(\omega)$ is the differential phase accumulated between the two local polarization eigenmodes, and $\hat{\mathbf{n}}_j$ identifies the local birefringence axis on the Bloch sphere.

The orientation of $\hat{\mathbf{n}}_j$ fixes which of the two orthogonal eigenmodes is associated with the positive eigenvalue of $\hat{\mathbf{n}}_j \cdot \boldsymbol{\sigma}$. In the convention used below, the slow mode corresponds to the $+1$ eigenvalue and the fast mode to the -1 eigenvalue. Reversing the orientation $\hat{\mathbf{n}}_j \to -\hat{\mathbf{n}}_j$ changes the sign in the exponential but leaves all physical predictions unchanged provided that the convention is used consistently.

At a fixed frequency, Eq. (2.98) describes a coherent polarization rotation. The connection between this general birefringent evolution and the reduced phase model introduced earlier is made explicit in the following subsection.

2.8.1. Fixed-Frequency Limit and Connection to the Residual-Phase Model

Consider first a monochromatic component at the carrier frequency ω_0. Let $|f_j\rangle$ and $|s_j\rangle$ denote the local fast and slow polarization eigenmodes of the segment, with propagation constants $\beta_{f,j}(\omega_0)$ and $\beta_{s,j}(\omega_0)$, respectively.

An arbitrary input polarization state can be expanded in this eigenmode basis as

$$|\chi_{\text{in}}\rangle = a|f_j\rangle + b|s_j\rangle. \tag{2.99}$$

Because the segment is uniform, the two eigenmodes propagate independently. After a distance L_j,

$$|\chi_{\text{out}}\rangle = a e^{i\beta_{f,j}(\omega_0)L_j}|f_j\rangle + b e^{i\beta_{s,j}(\omega_0)L_j}|s_j\rangle. \tag{2.100}$$

Introducing the mean and differential propagation constants,

$$\bar{\beta}_j = \frac{\beta_{s,j} + \beta_{f,j}}{2}, \qquad \Delta\beta_j = \beta_{s,j} - \beta_{f,j}, \tag{2.101}$$

the individual propagation constants can be written as

$$\beta_{f,j} = \bar{\beta}_j - \frac{\Delta\beta_j}{2}, \qquad \beta_{s,j} = \bar{\beta}_j + \frac{\Delta\beta_j}{2}.$$

Using the relation introduced in Eq. (2.94),

$$\Delta\phi_j(\omega_0) = L_j \Delta\beta_j(\omega_0), \tag{2.102}$$

the propagation operator in the local eigenmode basis $\{|f_j\rangle, |s_j\rangle\}$ becomes

$$U_j^{\text{eig}}(\omega_0) = e^{i\bar{\beta}_j L_j} \begin{pmatrix} e^{-i\Delta\phi_j(\omega_0)/2} & 0 \\ 0 & e^{+i\Delta\phi_j(\omega_0)/2} \end{pmatrix}. \tag{2.103}$$

The common factor $e^{i\bar{\beta}_j L_j}$ is a global phase and therefore does not affect polarization observables. The physically relevant transformation is consequently determined by the differential phase $\Delta\phi_j$.

For a generic birefringent segment, the eigenmodes need not coincide with the laboratory horizontal–vertical basis. Let

$$V_j = \begin{pmatrix} |f_j\rangle & |s_j\rangle \end{pmatrix} \tag{2.104}$$

denote the unitary matrix whose columns are the two eigenmodes expressed in the laboratory basis. The corresponding fixed-frequency Jones operator is

$$U_j(\omega_0) = e^{i\bar{\beta}_j L_j} V_j \begin{pmatrix} e^{-i\Delta\phi_j/2} & 0 \\ 0 & e^{+i\Delta\phi_j/2} \end{pmatrix} V_j^{\dagger}. \tag{2.105}$$

Equation (2.105) represents the generic fixed-frequency birefringence model. Apart from the polarization-independent phase, it is a unitary rotation around the local birefringence axis $\hat{\mathbf{n}}_j$. It is therefore equivalent to the fixed-frequency limit of Eq. (2.98).

A direct connection with the reduced phase operator can first be established through the special case in which the segment eigenmodes themselves coincide with the computational polarization basis,

$$|f_j\rangle \equiv |H\rangle \equiv |0\rangle, \qquad |s_j\rangle \equiv |V\rangle \equiv |1\rangle. \tag{2.106}$$

In this particular case $V_j = I$, and Eq. (2.103) reduces to

$$U_j(\omega_0) = e^{i\bar{\beta}_j L_j} \begin{pmatrix} e^{-i\Delta\phi_j/2} & 0 \\ 0 & e^{+i\Delta\phi_j/2} \end{pmatrix}. \tag{2.107}$$

Factoring out the common phase associated with the first component gives

$$U_j(\omega_0) = e^{i\left(\bar{\beta}_j L_j - \Delta\phi_j/2\right)} \begin{pmatrix} 1 & 0 \\ 0 & e^{i\Delta\phi_j} \end{pmatrix}. \tag{2.108}$$

Since the prefactor is physically irrelevant for polarization observables, the effective transformation is

$$U_j^{\mathrm{red}}(\theta_j) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta_j} \end{pmatrix}, \qquad \theta_j = \Delta\phi_j(\omega_0) = L_j \Delta\beta_j(\omega_0). \tag{2.109}$$

Equation (2.109) shows algebraically that the reduced phase operator is contained as a special fixed-frequency case of the general birefringent transformation. This aligned single-segment example, however, should not be interpreted as implying that polarization compensation aligns the local birefringence eigenmodes throughout a real fiber.

In a nonuniform fiber, the local fast and slow axes generally vary along the propagation direction. The fixed-frequency transformation of the complete fiber is instead obtained from the ordered product of the transformations of all its segments,

$$U_{\mathrm{fib}}(\omega_0) = U_N(\omega_0) U_{N-1}(\omega_0) \cdots U_1(\omega_0). \tag{2.110}$$

The polarization-compensation procedure employed in the experimental setting acts on this complete input–output transformation rather than on the internal local eigenmodes of the individual fiber segments. Within the reduced model adopted in this work, compensation realigns the relevant output polarization axis with the computational reference basis while leaving a residual rotation around that stabilized axis. The effective compensated transformation can therefore be represented, up to an overall phase, as

$$U_{\text{fib}}^{\text{comp}}(\omega_0) \sim \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}, \tag{2.111}$$

where θ represents the residual coherent phase not removed by the compensation procedure.

Thus, the reduced phase model is not a different physical propagation mechanism. Rather, it is a one-parameter effective description of the residual fixed-frequency transformation of the compensated fiber. The aligned single-segment case above provides the simplest mathematical realization of the same operator form, without requiring the local birefringence axes of a real segmented fiber to coincide with the laboratory H/V basis.

For the two-arm configuration considered in this work, let the corresponding residual phases be θ_A and θ_B. Starting from

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right),$$

the local transformation gives

$$\left[U_A^{\text{red}}(\theta_A) \otimes U_B^{\text{red}}(\theta_B)\right] |\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(e^{i\theta_B}|01\rangle + e^{i\theta_A}|10\rangle\right) \sim \frac{1}{\sqrt{2}}\left(|01\rangle + e^{i(\theta_A-\theta_B)}|10\rangle\right), \tag{2.112}$$

where $\sim$ denotes equality up to a global phase. Defining

$$\theta = \theta_A - \theta_B, \tag{2.113}$$

one recovers the one-parameter reduced phase state used in the preceding modeling levels.

This fixed-frequency transformation remains fully unitary. It can modify polarization correlations and measurement outcomes, but it cannot by itself reduce the purity of the polarization state. Polarization decoherence emerges only when the frequency dependence of the birefringent transformation becomes relevant over the finite spectral bandwidth of the photon.

2.8.2. First-Order Frequency Dependence and Polarization–Frequency Coupling

The fixed-frequency description can now be extended to a finite-bandwidth photon. Over the relevant spectral interval, the local birefringence axis is assumed to be frequency independent to first order,

$$\hat{\mathbf{n}}_j(\omega) \simeq \hat{\mathbf{n}}_j(\omega_0) \equiv \hat{\mathbf{n}}_j,$$

while the accumulated differential phase retains its frequency dependence.

It is convenient to introduce the angular-frequency offset from the carrier,

$$\Omega = \omega - \omega_0, \tag{2.114}$$

so that

$$\omega = \omega_0 + \Omega. \tag{2.115}$$

The carrier frequency therefore corresponds to $\Omega = 0$. This change of variable will also be adopted in the numerical implementation, where the spectral grids are constructed directly in terms of frequency offsets rather than absolute optical frequencies.

Expanding the differential phase around the carrier frequency gives

$$\Delta\phi_j(\omega_0 + \Omega) \simeq \Delta\phi_j(\omega_0) + \Omega\Delta\tau_j, \tag{2.116}$$

where

$$\Delta\tau_j = \left.\frac{d\Delta\phi_j}{d\omega}\right|_{\omega_0} \tag{2.117}$$

is the signed differential group-delay difference introduced in Section 2.7 [20]. Its magnitude $|\Delta\tau_j|$ is the physical differential group delay of the segment.

To isolate the polarization-dependent part of the propagation, consider Eq. (2.98) without the common polarization-independent phase,

$$\widetilde{U}_j(\omega) = \exp\left[\frac{i}{2}\Delta\phi_j(\omega)\hat{\mathbf{n}}_j \cdot \boldsymbol{\sigma}\right]. \tag{2.118}$$

Using the frequency-offset variable, the same physical operator can be parameterized as

$$\mathcal{U}_j(\Omega) \equiv \widetilde{U}_j(\omega_0 + \Omega). \tag{2.119}$$

Substituting the first-order expansion of Eq. (2.116) gives

$$\mathcal{U}_j(\Omega) \simeq \exp\left[\frac{i}{2}\left(\Delta\phi_j(\omega_0) + \Omega\Delta\tau_j\right)\hat{\mathbf{n}}_j \cdot \boldsymbol{\sigma}\right]. \tag{2.120}$$

Equations (2.118) and (2.120) describe the same transformation: the former uses the absolute optical frequency ω, whereas the latter uses the detuning $\Omega = \omega - \omega_0$.

A compensating transformation that removes the polarization evolution at the carrier frequency is

$$C_j = \widetilde{U}_j^{\dagger}(\omega_0) = \mathcal{U}_j^{\dagger}(0). \tag{2.121}$$

Since the birefringence axis is assumed constant over the bandwidth, the carrier transformation and the frequency-dependent transformation commute. The residual operator can therefore be written as

$$\begin{aligned}\mathcal{U}_j^{\mathrm{res}}(\Omega) &\equiv U_j^{\mathrm{res}}(\omega_0 + \Omega)\\ &= C_j\mathcal{U}_j(\Omega)\\ &\simeq \exp\left[\frac{i}{2}\Omega\Delta\tau_j\hat{\mathbf{n}}_j \cdot \boldsymbol{\sigma}\right].\end{aligned} \tag{2.122}$$

Thus, once the carrier-frequency transformation has been compensated, the first-order residual polarization evolution depends only on the frequency offset Ω, the signed differential delay $\Delta\tau_j$, and the local birefringence axis $\hat{\mathbf{n}}_j$.

The polarization-independent factor $e^{i\bar{\phi}_j(\omega)}$ has been omitted from Eq. (2.122). For each spectral component it acts identically on both polarization eigenmodes and therefore cancels from the reduced polarization density operator. Its frequency dependence may nevertheless describe polarization-independent temporal propagation and chromatic dispersion, which are outside the polarization-focused model considered here.

Let $|s_+\rangle$ and $|s_-\rangle$ denote the eigenstates of the local birefringence generator,

$$(\hat{\mathbf{n}}_j \cdot \boldsymbol{\sigma})\,|s_\pm\rangle = \pm|s_\pm\rangle. \tag{2.123}$$

An input polarization state can be decomposed as

$$|\chi\rangle = \alpha|s_+\rangle + \beta|s_-\rangle. \tag{2.124}$$

Let $f(\omega)$ denote the spectral amplitude of the photon. In terms of the frequency-offset coordinate, define

$$f_\Omega(\Omega) \equiv f(\omega_0 + \Omega). \tag{2.125}$$

Since $d\omega = d\Omega$, the normalization condition becomes

$$\int d\Omega\,|f_\Omega(\Omega)|^2 = 1. \tag{2.126}$$

If polarization and frequency are initially separable, the input wave packet may therefore be written as

$$|\Psi_{\mathrm{in}}\rangle = \int d\Omega\, f_\Omega(\Omega)|\omega_0 + \Omega\rangle \otimes |\chi\rangle. \tag{2.127}$$

Applying the residual operator of Eq. (2.122) gives

$$|\Psi_{\mathrm{out}}\rangle = \int d\Omega\, f_\Omega(\Omega)|\omega_0 + \Omega\rangle \otimes \left[\alpha e^{+i\Omega\Delta\tau_j/2}|s_+\rangle + \beta e^{-i\Omega\Delta\tau_j/2}|s_-\rangle\right]. \tag{2.128}$$

The complete polarization–frequency state remains pure because the propagation is unitary. Unless the input polarization coincides with one of the local birefringence eigenstates, however, the polarization and frequency degrees of freedom are no longer separable.

In the time-domain representation, the opposite frequency-dependent phases in Eq. (2.128) correspond to opposite temporal translations of the two eigenmode components. Their relative temporal separation is the differential group delay $|\Delta\tau_j|$.

This mechanism is illustrated schematically in Fig. 2.3.

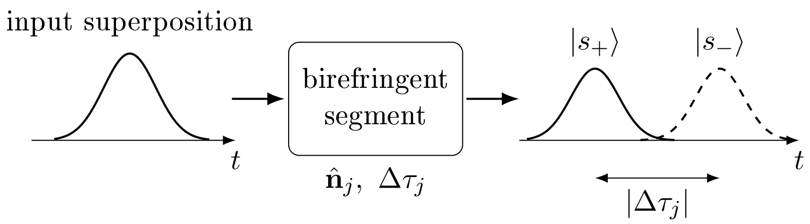


Figure 2.3: Schematic time-domain interpretation of first-order polarization-mode dispersion in a single birefringent segment. An input superposition of the two local polarization eigenmodes evolves into two wave-packet components separated by the differential group delay $|\Delta\tau_j|$. The complete polarization–frequency state remains pure, while reduced polarization coherence decreases when the temporal degree of freedom is not resolved and the overlap between the two components becomes incomplete.

If the spectral degree of freedom is not resolved by the measurement, the polarization state is obtained by tracing it out,

$$\rho_{\mathrm{pol}}^{\mathrm{out}} = \mathrm{Tr}_\omega \left(|\Psi_{\mathrm{out}}\rangle\langle\Psi_{\mathrm{out}}| \right). \tag{2.129}$$

Since the transformation $\omega = \omega_0 + \Omega$ is a simple translation of the spectral coordinate, the same partial trace may equivalently be evaluated over Ω. The diagonal polarization populations remain unchanged, whereas the off-diagonal coherence terms are multiplied by

$$\kappa_j = \int d\Omega\, |f_\Omega(\Omega)|^2 e^{i\Omega\Delta\tau_j}. \tag{2.130}$$

Consequently, in the local eigenmode basis,

$$\rho_{\mathrm{pol}}^{\mathrm{out}} = \begin{pmatrix} \rho_{++}^{\mathrm{in}} & \kappa_j \rho_{+-}^{\mathrm{in}} \\ \kappa_j^* \rho_{-+}^{\mathrm{in}} & \rho_{--}^{\mathrm{in}} \end{pmatrix}. \tag{2.131}$$

The factor κ_j is the characteristic function of the spectral probability density, expressed here in the frequency-offset coordinate and evaluated at the differential delay. Its magnitude therefore quantifies the residual indistinguishability, and hence the polarization coherence, of the two delayed eigenmode components [5].

In the monochromatic limit,

$$|f_\Omega(\Omega)|^2 \to \delta(\Omega),$$

one obtains

$$|\kappa_j| = 1.$$

The polarization state therefore remains fully coherent, consistently with the fixed-frequency unitary model derived in Section 2.8.1.

For a finite bandwidth, $|\kappa_j| < 1$ whenever the spectral averaging makes the two polarization components partially distinguishable in the unresolved temporal degree of freedom. The resulting loss of reduced polarization coherence does not originate from non-unitary evolution of the complete system; it arises from polarization–frequency coupling followed by a partial trace over frequency.

For a Gaussian spectral probability density expressed in terms of the frequency offset,

$$P_\Omega(\Omega) = |f_\Omega(\Omega)|^2 = \frac{1}{\sqrt{2\pi}\sigma_\omega} \exp\left[-\frac{\Omega^2}{2\sigma_\omega^2}\right], \tag{2.132}$$

where σ_ω is the root-mean-square width of the *spectral probability density*, Eq. (2.130) gives

$$|\kappa_j| = \exp\left[-\frac{1}{2}\sigma_\omega^2 \Delta\tau_j^2\right]. \tag{2.133}$$

The dimensionless product

$$\xi_j = \sigma_\omega |\Delta\tau_j|$$

therefore controls the strength of the polarization-coherence reduction. The same differential group delay has little influence on a sufficiently narrowband photon, whereas it can strongly suppress the reduced polarization coherence when the spectral bandwidth becomes sufficiently large [5, 6].

This offset-frequency formulation is also the one adopted in the MATLAB implementation. The numerical routines discretize Ω around zero and evaluate the operator

$\mathcal{U}_j(\Omega) = \widetilde{U}_j(\omega_0 + \Omega)$. The use of Ω therefore represents only a change of spectral coordinate and does not introduce an additional physical approximation beyond the first-order expansion of the differential phase.

This single-segment description provides the elementary physical building block for the concatenated birefringent-fiber model developed in the following section.

2.9. Concatenated Birefringent Fiber Model

The local birefringence of a real optical fiber generally varies along the propagation direction because of geometrical imperfections, residual stress, bending, twisting, and environmental perturbations. A nonuniform fiber can therefore be modeled as a sequence of locally uniform birefringent segments [7, 8, 19].

Figure 2.4 illustrates this representation.

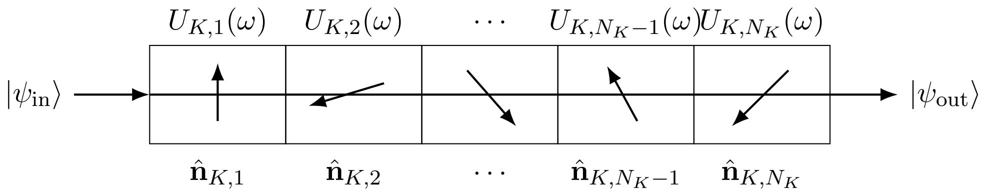


Figure 2.4: Segmented representation of a nonuniform birefringent fiber. Each section is characterized by its own birefringence axis and frequency-dependent unitary operator. The different axis orientations schematically represent the spatial variation of local birefringence along the fiber. The total fiber transformation is the ordered product of the individual segment operators.

For arm K, let N_K denote the number of segments. If the photon encounters segment 1 first and segment N_K last, the complete transformation is

$$U_K(\omega_K) = U_{K,N_K}(\omega_K) \cdots U_{K,2}(\omega_K) U_{K,1}(\omega_K). \tag{2.134}$$

The spatial ordering is essential because transformations associated with different birefringence axes generally do not commute:

$$U_{K,j}(\omega) U_{K,k}(\omega) \neq U_{K,k}(\omega) U_{K,j}(\omega). \tag{2.135}$$

Each segment therefore changes the polarization state on which all subsequent segments act.

If all segments share the same oriented birefringence axis, the operators commute and the differential phases add directly. In this special case, the complete fiber reduces to an equivalent single birefringent element. In the general case, the complete ordered product must be retained.

2.9.1. Fixed and Statistical Fiber Realizations

A particular segmented fiber is defined by the collection of parameters associated with its constituent sections. A realization of arm K may be represented schematically as

$$\mathcal{R}_K = \{\hat{\mathbf{n}}_{K,j}, \Delta\phi_{K,j}(\omega_{0,K}), \Delta\tau_{K,j}\}_{j=1}^{N_K}. \tag{2.136}$$

For a fixed realization, all of these quantities are prescribed and the frequency-dependent operator $U_K(\omega)$ is deterministic.

A statistical fiber model instead samples some or all segment parameters from specified probability distributions. Such sampling represents uncertainty or variation in the spatial birefringence profile. The particular statistical distributions and numerical discretization adopted in the simulator are introduced in the numerical framework chapter.

It is important to distinguish spatial randomness from quantum-state mixedness. For one fixed fiber realization and one resolved frequency pair, propagation through the entire segmented fiber remains unitary. Random orientation of the segments along the fiber does not by itself make the propagated state mixed.

Mixedness in the reduced polarization state can instead arise from:

- tracing over unresolved spectral or temporal degrees of freedom;
- statistical averaging over different fiber realizations if the physical fiber changes during the observation interval.

These two averaging procedures describe different physical mechanisms and are therefore kept conceptually distinct throughout the model.

2.10. Principal States of Polarization and Differential Group Delay

The local birefringence eigenmodes of the individual segments do not generally coincide with the principal states of polarization (PSPs) of the complete fiber. The effective first-order PMD properties must instead be extracted from the frequency dependence of the complete ordered Jones operator [7, 19].

The complete operator is decomposed into a polarization-independent phase and an SU(2) polarization transformation,

$$U_K(\omega) = e^{i\Phi_K(\omega)} W_K(\omega), \qquad W_K(\omega) \in \mathrm{SU}(2). \tag{2.137}$$

The derivative of the scalar phase $\Phi_K(\omega)$ represents a group-delay contribution common to both polarization components. It does not affect the reduced polarization state, although it may influence the absolute arrival time of the optical wave packet.

The polarization-dependent first-order response is described by the input-side PMD generator

$$G_K^{\mathrm{in}}(\omega) = -iW_K^\dagger(\omega)\frac{\partial W_K(\omega)}{\partial \omega}. \tag{2.138}$$

Because W_K is unitary with unit determinant, G_K^{in} is Hermitian and traceless. It can therefore be expanded in the Pauli basis as

$$G_K^{\mathrm{in}} = \frac{1}{2}\boldsymbol{\tau}_K \cdot \boldsymbol{\sigma}, \tag{2.139}$$

where $\boldsymbol{\tau}_K$ is the PMD vector.

Its magnitude defines the non-negative first-order differential group delay,

$$\tau_{\mathrm{DGD},K} = \|\boldsymbol{\tau}_K\|. \tag{2.140}$$

Equivalently, the eigenvalues of G_K^{in} are

$$\lambda_\pm = \pm\frac{\tau_{\mathrm{DGD},K}}{2}. \tag{2.141}$$

The corresponding eigenvectors,

$$|p_{K,+}^{\text{in}}\rangle, \qquad |p_{K,-}^{\text{in}}\rangle, \tag{2.142}$$

define the two input principal states of polarization. The corresponding output PSPs at the carrier frequency are

$$|p_{K,\pm}^{\text{out}}\rangle = W_K(\omega_{0,K})|p_{K,\pm}^{\text{in}}\rangle. \tag{2.143}$$

To first order around the carrier frequency,

$$W_K(\omega) \simeq W_K(\omega_{0,K}) \exp\left[i(\omega - \omega_{0,K})G_K^{\text{in}}(\omega_{0,K})\right]. \tag{2.144}$$

A principal state therefore propagates according to

$$W_K(\omega)|p_{K,\pm}^{\text{in}}\rangle \simeq e^{\pm i(\omega-\omega_{0,K})\tau_{\text{DGD},K}/2}|p_{K,\pm}^{\text{out}}\rangle. \tag{2.145}$$

Thus, within the first-order PMD approximation, a photon launched in one PSP emerges with a frequency-independent output polarization. The spectral components differ only by a frequency-dependent phase associated with an early or late temporal displacement.

A superposition of the two PSPs can instead become correlated with arrival time, providing the physical mechanism through which PMD reduces polarization coherence after the unresolved temporal or spectral degree of freedom is traced out.

The PSPs of the complete fiber must not be confused with the local birefringence eigenmodes of individual segments. Local eigenmodes diagonalize the Jones operator of one fiber section, whereas the PSPs diagonalize the first-order frequency response of the complete concatenated transformation.

The direction assigned to the PMD vector depends on the phase and Stokes-space conventions adopted in the literature. The physically invariant quantities used here are the PSP axis and the non-negative DGD $\tau_{\text{DGD},K}$.

2.11. PMD Acting on Polarization-Entangled Photon Pairs

For the two-arm system, the polarization transformation associated with a resolved frequency pair is

$$U_{AB}(\omega_A, \omega_B) = U_A(\omega_A) \otimes U_B(\omega_B). \tag{2.146}$$

The observable polarization state after the unresolved spectral degrees of freedom have been traced out is therefore

$$\rho_{AB}^{\text{pol,out}} = \iint d\omega_A \, d\omega_B \, J(\omega_A, \omega_B) \left[U_A(\omega_A) \otimes U_B(\omega_B)\right] \rho_{AB}^{\text{pol,in}} \left[U_A(\omega_A) \otimes U_B(\omega_B)\right]^\dagger . \tag{2.147}$$

For every individual frequency pair, the polarization transformation remains a product of local unitary operators. The effective reduced polarization map can nevertheless be non-unitary because different spectral components undergo different transformations and are subsequently combined through the spectral trace [5–7, 9].

If the joint spectral intensity factorizes,

$$J(\omega_A, \omega_B) = J_A(\omega_A) J_B(\omega_B), \tag{2.148}$$

the reduced polarization transformation factorizes into independently averaged local channels,

$$\mathcal{E}_{AB} = \mathcal{E}_A \otimes \mathcal{E}_B. \tag{2.149}$$

If the photon frequencies are spectrally correlated, however,

$$J(\omega_A, \omega_B) \neq J_A(\omega_A) J_B(\omega_B), \tag{2.150}$$

the local transformations in the two arms are sampled according to a correlated joint distribution.

The physical action remains local for each frequency pair because

$$U_{AB}(\omega_A, \omega_B) = U_A(\omega_A) \otimes U_B(\omega_B).$$

Nevertheless, the effective polarization map does not generally factorize into two independently averaged local channels. The spectral correlations inherited from the SPDC source therefore induce correlations between the effective polarization noise experienced in the two arms.

This distinction does not imply a physical interaction between the remote photons. It reflects instead the fact that the local frequency-dependent transformations are averaged according to a common correlated spectral distribution.

PMD thus provides a physical realization of the mechanism introduced conceptually earlier in the chapter: an underlying lossless and unitary polarization–frequency evolution can appear as polarization decoherence when the spectral and temporal degrees of freedom are not observed.

2.12. Relation Between the Modeling Levels

The models introduced in this chapter are not competing descriptions of the same propagation process. They form the progressive hierarchy summarized in Fig. 2.1, with different physical information retained at each level.

The fixed relative-phase model isolates coherent polarization evolution after partial compensation. Because the transformation is local and unitary, it changes the orientation of measured correlations without degrading purity or entanglement.

The statistical phase model introduces unresolved variability of the coherent transformation. Ensemble averaging suppresses transverse coherence through the complex parameter

$$\mu = \left\langle e^{i\theta} \right\rangle . \tag{2.151}$$

The magnitude $|\mu|$ quantifies coherence loss, while its phase retains the remaining coherent orientation.

The global depolarizing model provides an isotropic analytical baseline. The local two-arm depolarizing model adapts this description to the physical geometry of separate propagation paths through the parameters p_A and p_B. Their combined action on bipartite correlations is summarized by

$$\eta = (1 - p_A)(1 - p_B). \tag{2.152}$$

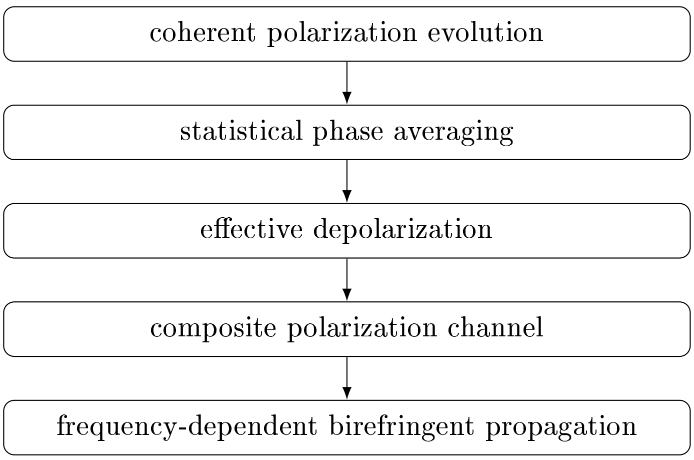
The composite effective model combines coherent phase evolution, statistical dephasing, and local isotropic depolarization within a common polarization-only framework.

These effective models remain useful because they provide closed-form analytical predictions, make different degradation signatures transparent, and constitute controlled benchmarks for validating the numerical implementation. They should not, however, be interpreted as complete microscopic models of optical-fiber propagation.

The frequency-dependent framework extends the state space by retaining the spectral degrees of freedom. Local birefringence produces a frequency-dependent phase, whose derivative gives a differential group delay. For nonuniform fibers, the complete response is obtained through an ordered concatenation of birefringent sections, and the principal states and total DGD emerge from the frequency derivative of the complete Jones operator.

In this final description, polarization decoherence is not imposed directly through a phenomenological noise probability. For a fixed lossless fiber realization, the enlarged polarization–frequency state evolves unitarily. Effective polarization mixedness emerges from polarization–frequency coupling followed by a partial trace over spectral degrees of freedom that are not resolved experimentally.

The chapter therefore establishes a continuous modeling path from simple analytically controlled channels to a physical PMD description:



This hierarchy provides the theoretical basis for Chapter **??**. The analytically tractable models are used first to validate the state-evolution, channel, correlation, and metric routines. The frequency-dependent model is subsequently implemented through spectral discretization and ordered propagation across birefringent fiber segments, allowing polarization degradation to be computed from the physical model rather than prescribed phenomenologically.

3 | Numerical Experiments and Results

The theoretical framework developed in the previous chapters is implemented and investigated numerically in this chapter. The objective is to move from the analytical description of polarization-entangled photon propagation to a computational framework capable of quantifying how physically relevant fiber and source parameters affect the distributed quantum state.

The numerical analysis is organized around a progressive increase in physical complexity. Controlled limiting cases are considered first in order to validate the implementation against analytical predictions. The study is then extended to frequency-dependent birefringent propagation, two-arm polarization-mode dispersion, and statistical ensembles of randomly birefringent fibers. Particular attention is given to the dependence of the output state on quantities that can be related directly to the physical system, including photon bandwidth, differential group delay, fiber length, birefringence statistics, principal-state orientations, and the spectral correlations of the photon pair.

The purpose of the numerical experiments is therefore not limited to verifying the correctness of the implementation. The simulator is used as an analysis tool to identify the physical regimes in which fiber propagation produces negligible, partial, or substantial degradation of polarization entanglement, and to determine how these regimes are reflected in experimentally accessible quantities. The output states are characterized through polarization correlations, reduced density operators, purity, fidelity, concurrence, and Bell-nonlocality indicators. When the fiber model is stochastic, these quantities are treated statistically over an ensemble of fiber realizations rather than solely through their mean values.

A further objective is to establish a direct connection between the numerical model and the experimental investigation developed later in this work. Whenever possible, the numerical experiments are therefore formulated in terms of parameters that can either be controlled or estimated experimentally. This allows the simulator to provide reference predictions for the interpretation of measured single-photon and coincidence statistics and, more generally, to investigate the range of fiber and source conditions compatible with the preservation of useful polarization entanglement.

3.1. Computational Framework and MATLAB Library

All numerical models presented in this chapter are implemented in MATLAB. This choice allows the simulator to be integrated directly into the pre-existing computational codebase used throughout the project, avoiding the introduction of a separate software environment and facilitating maintenance, validation, and future extensions of the framework.

MATLAB is also particularly well suited to the mathematical structure of the problem considered in this work. Quantum states, density operators, Jones operators, Pauli observables, tensor products, and frequency-dependent propagation matrices are represented directly through complex vectors and matrices. As a consequence, the numerical objects used in the implementation closely reproduce both the notation and the algebraic structure of their theoretical counterparts. For example, single-photon polarization operators are represented by 2×2 matrices, bipartite states and operators by vectors and matrices in $\mathbb{C}^2 \otimes \mathbb{C}^2$, and local bipartite transformations are constructed explicitly through tensor products.

The resulting implementation is organized as a modular MATLAB library rather than as a collection of experiment-specific scripts. Reusable routines are provided for state preparation, operator construction, local and frequency-dependent propagation, quantum-channel application, partial traces, correlation evaluation, state metrics, spectral integration, segmented-fiber construction, statistical sampling, and numerical validation. Higher-level numerical experiments then combine these validated components without redefining the underlying mathematical operations.

This architecture provides a direct correspondence between the theoretical model and its computational realization:

Physical and Mathematical Model
$\downarrow$
Reusable MATLAB Operators and State Routines
$\downarrow$
Frequency-Resolved Fiber Propagation
$\downarrow$
Spectral and Statistical Averaging
$\downarrow$
Density Operator and Observable Evaluation
$\downarrow$
Numerical Experiments and Experimental Predictions.

Separating the reusable numerical library from the individual experiment workflows has two main advantages. First, the same implementation of each mathematical operation is used throughout the thesis, reducing duplication and making numerical consistency easier to verify. Second, progressively more complex fiber models can be constructed by composition of previously validated components, following the same hierarchical strategy adopted in the theoretical development.

The numerical study presented in the following sections therefore proceeds from controlled validation cases toward increasingly realistic and statistically characterized models of fiber propagation.

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